

Back to Basics: Basic Research Spillovers, Innovation Policy and Growth*

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Abstract

This paper introduces a general equilibrium model of endogenous technical change through basic and applied research. Basic research differs from applied research in the nature and the magnitude of the generated spillovers. We propose a novel way of empirically identifying these spillovers and embed them in a framework with private firms and a public research sector. After characterizing the equilibrium, we estimate our model using micro-level data on research expenditures by French firms. Our key finding is that uniform research subsidies can accentuate the dynamic misallocation in the economy by oversubsidizing applied research. Policies geared towards public basic research and its interaction with the private sector are significantly welfare-improving.

Keywords: Productivity, innovation, basic research, applied research, research and development, government spending, endogenous growth, spillover.

JEL classification: J82, L25, L50, O31, O38, O40

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1 Introduction

Fostering innovation is one of the primary objectives of economists and policymakers. The amount of resources invested in research is often at the heart of the debate over how to best achieve such a goal. Less well known, however, is what role the composition of research plays, particularly when considering the breakdown between basic and applied research. In this paper, we aim to fill this gap by studying the differential effects of basic versus applied research on aggregate innovation.

The distinction between basic and applied research is economically and empirically important. According to the National Science Foundation (NSF), basic research investment refers to a “systematic study to gain more comprehensive knowledge or understanding of the subject under study without specific applications in mind.” Conversely, applied research is defined as a “systematic study to gain knowledge or understanding to meet a specific, recognized need.” This distinction is empirically important since almost half of total research investment is allocated to basic research in countries such as France and the US.¹

In a recent report by the US Congress’s Joint Economic Committee, the issue of investment in basic research received fresh policy interest: the report argues that, despite its value to society as a whole, basic research is underfunded by private firms precisely because it is performed with no specific commercial applications in mind. It also states that the level of federal funding for basic research is “worrisome” and should be increased (Committee, 2016). Similarly, the NSF, the Sloan Foundation, and the National Bureau of Economic Research (NBER) have been proactively trying to draw attention to the declining government involvement in basic research funding.²

Despite clear empirical importance and considerable policy interest, the differential roles of basic and applied research in the growth process are still relatively unexplored from the macroeconomic perspective, and many related questions remain to be answered. How sizable are the spillovers from basic and applied research? What are the potential inefficiencies in a competitive economy? What are the appropriate government policies to mitigate these inefficiencies, and does the academic sector play a special role in determining the pace of innovation in the economy? This paper attempts to answer these important questions.

In order to understand the potential inefficiencies in research investment and to design appropriate industrial policies to address them, it is necessary to adopt a structural framework that explicitly models the incentives of private and public entities to engage in the different types of research. We propose a general equilibrium, multi-industry framework with private firms and a public research sector. In our model, basic research has two distinct features relative to applied research. First, as Nelson (1959) argues, “Successful basic research projects very often have practical value in many fields.” In our setting, basic research can, therefore, generate spillovers that affect subsequent innovations both within and across industries. Applied research, on the other hand, generates innovations

¹The OECD provides a breakdown of gross domestic expenditure on R&D expenditures by type of research: https://stats.oecd.org/Index.aspx?DataSetCode=GERD_TORD#.

²For instance, the NBER started a new initiative entitled the “The Science of Science Funding Initiative” (<http://admin.nber.org/drupal/SOSF>) funded by the Sloan Foundation.

within a targeted industry. Second, the potential returns from basic research depend on whether it was generated in the private or the public sector. In line with the “Ivory Tower” theory of academic research, basic research by private firms in our model will result in consumer products faster than that undertaken by public research labs.

These features of our model also reflect the important debate in science policy about the characteristics of basic and applied research as framed by [Stokes \(1997\)](#) in “Pasteur’s quadrant: Basic Science and Technological Innovation.” The academic sector represents pure scientific discovery without application in mind, best exemplified by Bohr’s exploration of the structure of the atom. At the other extreme, lies the purely use-based –i.e., applied– private research, such as Edison’s notable efforts in his Menlo Park laboratories. Finally, cutting across those extreme lines is “Pasteur’s Quadrant,” for work that is directly influenced in its course by the quest for fundamental understanding but inspired by an applied problem.³ In our model, this role will be filled by private basic research. Prominent examples include the work by Nobel prize winners Arno Penzias and Robert Wilson at Bell Labs, doing, according to [Rosenberg \(1990\)](#), “basic research in astrophysics because of its relationship to the whole field of problems and possibilities in microwave transmissions, and especially the use of communication satellites for such purposes.” Similarly, [Nelson \(1959\)](#) points out, “Carothers’s famous work in linear superpolymers began as an unrestricted foray into the unknown with no particular practical objective in view. But the research was in a new field of chemistry, and Du Pont believed that any new chemical breakthrough would probably be of value to the company.” In time, Carothers’ research led to the discovery and use of nylon by Du Pont.

It is important to note that, given the novelty of our theoretical framework, the policy implications are not widely known. At its core, our theoretical framework builds upon the work of [Klette and Kortum \(2004\)](#), yet we extend it substantially in a variety of directions. Most importantly, we allow for the distinction between basic and applied research, as well as public and private research. These unique features allow us to be the first to simultaneously discuss the roles that spillovers generated by different research types and within different institutional arrangements, such as firms and universities, play for economic growth. It allows us to understand under what alternative specifications policy makers pushing for more public funding for private research could be right.

Our ultimate goal in this paper is to undertake a quantitative investigation of the impact of various innovation policies on the aggregate economy. To this end, we estimate the structural parameters of our model using French firm-level data for the period 2000-2006. Information about research investment comes from the annual R&D Survey conducted by the French Ministry of Research. The advantage of these data is that they include information on the basic and applied research expenditures of individual firms. To measure the breadth of a firm’s activities, we combine two datasets (“Enquete Liaisons Financieres” (LIFI); “Enquete Annuelle des Entreprises” (EAE)) that allow us to precisely identify a firm’s links to different industries not only through product lines within the same

³The American physicist Gerald Holton called it “work that locates the center of research in an area of basic scientific ignorance that lies at the heart of a social problem.”

firm, but also through their ownership links. Finally, we use the balance sheet data from these firms to measure firm dynamics and NBER patent citation data to measure the quality of innovation. The final sample is composed of 13,708 firm-year observations.

These features of our data allow us to estimate the key spillovers involved in research. First, we measure the cross-industry spillovers associated with basic research through the investment choices of multi-industry firms. According to Nelson (1959), “a broad technological base insures that, whatever direction the path of research may take, the results are likely to be of value to the sponsoring firm.”⁴ In our data, we confirm that the proportion of basic research investment increases as the scope of a firm’s activities expands. In our analysis, we provide extensive robustness checks of our estimates with respect to confounding factors arising from firm heterogeneity, industry heterogeneity, and measurement error. Second, we use citation information from the patent data to empirically measure the quality (step size) for basic and applied innovation, as well as their importance for follow-up innovations (Hall, Jaffe, and Trajtenberg, 2001). Note that, to ensure an exact correspondence between theory and empirics, we extend our theoretical framework to incorporate patent citations. In particular, in the model, patents are generated through successful research and subsequently cited by follow-up patents. Finally, to further inform the model parameters on firm dynamics, we simultaneously target a rich set of firm characteristics related to multi-industry presence, profitability, age, and entry/exit patterns. We discuss their identification through a set of comparative statics and untargted moments.

Our main result is the quantification of the inefficiencies due to dynamic misallocation in research. We find that 68% of spillovers from basic research across industries are not internalized. As a result, there is a dynamic misallocation of research efforts, which reduces welfare by 4.6 percentage points in consumption-equivalent terms. One striking feature of the solution to the social planner’s problem is that the fraction of resources devoted to research activities is not substantially greater than in the decentralized equilibrium. Indeed, the dominant misallocation here is not that between production and research, as is common in this class of models, but among the various types of research activities, in this case, applied and basic innovation. Another striking feature is that, in the case of applied innovation, there is actually an *overinvestment* in the decentralized economy due to the strategic complementarity between basic research spillovers and the returns to applied research.

This raises an important question: to what extent can public policies address this inefficiency? The first policy we analyze is a uniform research subsidy to private firms. In this environment, subsidizing overall private research is ineffective since this policy exacerbates the existing overinvestment in applied research arising from excessive competition. Therefore, the welfare improvements from such a subsidy are limited unless the policymaker is able to discriminate between types of research projects at the firm level. Thus, we consider a type-dependent research subsidy and find that the optimal policy is to subsidize basic research by 49% and applied research by 11%. While the type-dependent subsidy is a promising policy tool to increase welfare, one concern is that it might be difficult to implement. As Rosenberg (1990) notes, it is not always clear whether or not a specific undertak-

⁴Specific examples include Gibb’s law of phases, which has been applied to activities ranging from petroleum refining to metal ore separation; or the development of the laser in 1964, which was used progressively in printers (1969), fiber optic communications (1970), bar code scanners (1974), vision correction surgery (1987), and DVD players (1995).

ing should be considered basic research. This might lead firms to misclassify research investment to exploit the differential subsidy rates. Therefore, we compute the optimal type-dependent policy under different degrees of misclassification. Importantly for policymakers, the welfare gains arising from a type-dependent subsidy are robust to a substantial amount of misclassification. Loosely speaking, the welfare gains associated with a type-dependent research subsidy are robust for levels of misclassification below 50%.

The appendix features an extensive set of theoretical and quantitative robustness checks. We re-estimate and evaluate the policy implications of the model with respect to industry heterogeneity and persistent firm types. We also re-estimate the model eliminating private basic research but maintaining public basic research; and eliminating both private and public basic research altogether. We then consider robustness checks with respect to the diffusion of public basic research and public research productivity. A final set of robustness checks relaxes assumptions about the spillover structure of applied research, and reevaluates the identification of the Constant Relative Risk Aversion (CRRA) parameter.

Related Literature Our main contribution is to the macro literature on endogenous technical change. Although policymakers have considered the different characteristics of basic and applied research, as well as public and private research, to be of first-order importance, these issues have received insufficient attention from the economic growth literature. In particular, models of endogenous technological change (see [Aghion, Akcigit, and Howitt \(2014\)](#) for a survey) mainly consider a uniform type of (applied) research and ignore basic research investment in the economy. A few exceptions are [Aghion and Howitt \(1996, 2009\)](#), [Cozzi and Galli \(2009, 2014\)](#), [Gersbach, Schneider, and Schneller \(2013\)](#), [Gersbach and Schneider \(2015\)](#), [Morales \(2004\)](#), and [Mansfield \(1995\)](#), all of whom consider theoretical models with both basic and applied research investment.

We contribute to this mostly-theoretical literature in various ways. First, we build a model with rich “firm dynamics” that is estimated with new firm-level micro data on firms’ research investment.⁵ To the best of our knowledge, ours is the first study to map a model with basic and applied research to firm-level data. In addition to including the private investment in basic research, we enrich the analysis of the distinct features of basic research by introducing a novel method to identify within- and cross-industry spillovers. Second, ours is the first framework that allows for a distinction between basic and applied research by private firms. Thanks to this feature, we are able to allow for differential efficiency properties of private basic versus applied research investments. One of our main results shows that innovation policies such as uniform research subsidies are ineffective since they oversubsidize applied research and undersubsidize basic research. Third, we complement this setting by also considering public research labs. This feature allows us to analyze the potential substitutability or complementarity between public and private research investments.

The paper also relates to the micro literature on innovation policy that discusses and quantifies the

⁵See Section 1.4 in [Aghion, Akcigit, and Howitt \(2014\)](#) for a detailed account of “realistic firm dynamics” in endogenous growth.

distinction between basic and applied research. Consistent with our framework, [Nelson \(1959\)](#) argues that a key distinction between these two types of research lies in the breadth of the potential discoveries. At the same time, [Rosenberg \(1990\)](#) provides a skeptical view on the possibility of categorizing research on the basis of the motives of the person performing it. We consider the associated measurement issue from within the context of our model and quantitative results. [Cohen and Klepper \(1992\)](#) focus on the relationship between firm size and R&D. In particular they argue that “social advantages of large firm size stem from the idea that large firms possess an advantage in appropriating the returns from innovation, due, in part, to imperfections in the market for information. This causes the returns to large firms to conduct more socially desirable R&D than smaller firms.” In this paper, we identify the economic scope of a firm as a factor attenuating the appropriability problem of research, and we conceptually disentangle it from a pure size effect. Empirically, several important papers ([Mansfield, 1980](#); [Mansfield, 1981](#); [Link, 1981](#); [Griliches, 1986](#)) show that the distinction between basic and applied research is quantitatively important. These papers typically evaluate the importance of basic and applied research by relating it to production data and estimating its output elasticity or rate of return from an extended Cobb-Douglas production function. The common conclusion from those papers is that basic research seems to command a significant “premium”—i.e., that its contribution to firm productivity is significantly larger than that of applied research.

Methodologically, our paper is related to the growing branch of endogenous growth with firm dynamics that estimates these models structurally using micro data. For instance, [Lentz and Mortensen \(2008, 2016\)](#) use a panel of Danish firms; [Acemoglu, Akcigit, Alp, Bloom, and Kerr \(2018\)](#), [Akcigit and Kerr \(2018\)](#), and [Garcia-Macia, Hsieh, and Klenow \(2019\)](#) use US Census of Manufacturing; [Peters \(2015\)](#) uses Indonesian micro data; and [Ates and Saffie \(2014\)](#) use Chilean firm-level data to estimate enriched versions of the quality-ladder models. A number of papers also study the role of innovation policy in a similar class of models. For instance, [Atkeson and Burstein \(2019\)](#) study the impact of policy-induced changes in innovative investment by firms on growth in aggregate productivity. [Akcigit, Hanley, and Stantcheva \(2016\)](#) use a mechanism design approach to solve for the optimal design of innovation policy using a quality-ladder model with asymmetric information on firm types. Finally, [Garicano, Lelarge, and Van Reenen \(2016\)](#) use firm-level data to study the impact of size-dependent policies on misallocation of factors of production in France. These papers do not consider the distinction between basic and applied research, whereas our focus is on estimating the associated spillovers using French firm-level data and design-relevant policies around these spillovers.

Consistent with our results, some papers show that the speed of technology diffusion is linked to the ability of inventors to utilize ideas for production ([Akcigit, Celik, and Greenwood \(2016\)](#)), or to the existing patent rights over those technologies ([Galasso and Schankerman \(2015\)](#)). In the same spirit, [Bloom, Schankerman, and Van Reenen \(2013\)](#) identify technology and product market spillovers for US manufacturing firms. They show that small firms generate lower social returns to R&D because they operate in technological niches. Our paper suggests that their result can be rationalized by these firms’ lower incentives to invest in basic research that is more difficult to appropriate. In our framework, more public basic research investment stimulates more private applied research.

In that regard, our results parallel the findings of [Cozzi and Impullitti \(2010\)](#) and [Impullitti \(2010\)](#), who show that increases in the technological content of public spending encourages more private R&D spending in the US.

The rest of the paper is organized as follows. In Section 2, we introduce the model, characterize its dynamic equilibrium, and discuss the main mechanisms. In Section 3, we present and discuss the empirical patterns that provide the basis for the structural estimation. Section 4 describes the estimation and identification of the model. Section 5 provides a detailed discussion of the welfare effects of various policies on the decentralized economy. Section 6 concludes.

2 Theory: Growth with Basic and Applied Research

Our theoretical framework will depart from standard endogenous growth models in a number of ways. First, we introduce a distinction in the appropriability of innovations from basic and applied research. Following the influential literature on basic science, we consider the possibility that basic research generates not only spillovers within an industry, but also across industries ([Nelson \(1959\)](#)). Thus, we model an economy in which firms can operate in multiple industries, a feature that endogenously generates incentives for firms to invest in basic research.

A second key feature of our model is the distinction between embodied and disembodied knowledge in the economy. Both private firms and public research labs are investing in basic research in this economy. However, successful basic research in the private sector is more likely to be immediately turned into a consumer product (embodied), as opposed to simply increasing the stock of knowledge available for future innovators (disembodied). This will induce a delay in the effect of public basic research. This *ivory tower* nature of academic research has been widely discussed in academic and policy circles, with a formal analysis being provided by [Aghion, Dewatripont, and Stein \(2008\)](#).⁶

2.1 Basic Environment

2.1.1 Production

Production is divided into three major levels: *upstream*, *midstream*, and *downstream*. First, the upstream level produces intermediate goods (y_{ij}); next, these goods are used to produce industry aggregates (Y_i) in the midstream level; and, finally, the downstream level combines these industry aggregates into the final good (Z). We will now describe them in detail.

Downstream Production The final good $Z(t)$ is produced in the downstream by infinitely many competitive firms that combine inputs from M different industries according to the following Cobb-Douglas production function:

$$Z(t) = \prod_{i=1}^M Y_i(t)^{\frac{1}{M}}. \quad (1)$$

⁶In their model, the academic sector is a precommitment mechanism that allows scientists to freely pursue their own interests. Consistent with our model, academic scientists may, therefore, end up working on projects with little immediate economic value. An important difference, however, is that in their setup, there is full appropriability of the innovation. We relax this assumption and show that it generates a delay in the innovation process with consequences for economic growth.

In this production function, $Y_i(t)$ is the aggregate output from industry $i \in \{1, \dots, M\}$. The economy consists of $M \in \mathbb{N}^+$ industries. In the context of firm-level data, each industry i can be thought of as a different 1-digit Standard Industrial Classification (SIC) code and $Z(t)$ is simply the aggregate GDP of the economy.⁷ We normalize the price of the final good to 1 at every instant t without any loss of generality. For notational simplicity, time subscripts will henceforth be suppressed.

Midstream Production Each industry aggregate Y_i is produced competitively, combining inputs from a continuum of product lines. Let y_{ij} denote the production of upstream good j in industry i by the firm that has the best technology in that product line. Industry aggregate i is produced according to the following CES production function:

$$Y_i = \left[\int_0^1 y_{ij}^{\frac{\varepsilon-1}{\varepsilon}} dj \right]^{\frac{\varepsilon}{\varepsilon-1}}. \quad (2)$$

Upstream Production In product line j , the firm that has the latest (and also the best) technology produces as a monopolist according to the following linear production technology that takes only labor as an input:

$$y_{ij} = q_{ij} l_{ij}, \quad (3)$$

where $q_{ij} > 0$ is the labor productivity associated with product line j , and l_{ij} is the number of production workers employed. Let us denote the wage rate in the economy by w in terms of the final good. The specification in (3) implies that each product y_{ij} has a constant marginal cost of production $w/q_{ij} > 0$. We denote the productivity index of industry i by

$$\bar{q}_i \equiv \left(\int_0^1 q_{ij}^{\varepsilon-1} dj \right)^{\frac{1}{\varepsilon-1}}. \quad (4)$$

Definition of a Firm In this model, as in Klette and Kortum (2004), a firm is defined as a collection of product lines in which it is the lead producer. These product lines can come from multiple industries. In what follows, $m_f \in \{1, \dots, M\}$ will denote the number of industries in which the firm actively operates; $n_{if} \in \mathbb{N}^0$ will denote the number of product lines firm f owns in a given industry i (e.g., $n_{if} = 0$ means no presence of firm f in industry i); and n_f will stand for the total number of product lines owned by the firm and will satisfy $n_f \equiv \sum_{i=1}^M n_{if}$. Henceforth, for notational tractability, we will drop the firm index f , when it creates no confusion.

A firm's payoff in a given product line j in industry i depends on its productivity level q_{ij} . Therefore, the payoff-relevant state of a firm is denoted by

$$\mathbf{q} = (q_1, q_2, \dots, q_m),$$

⁷Note that we introduce this multi-industry structure in order to model cross-industry spillovers. To avoid any additional theoretical complications, we will focus on symmetric equilibria in which industry aggregates assume a common value.

where $\mathbf{q}_i = \{q_{i,j_1}, q_{i,j_2}, \dots, q_{i,j_{n_i}}\}$ is a multi-set keeping track of all the productivity levels of the firm in industry i , where it has the best technology.⁸

Example 1. An example is helpful to summarize the description thus far. Figure 1 illustrates an economy that consists of $M = 4$ industries and a firm that operates in $m = 3$ industries ($i = 1, i = 3$, and $i = 4$). It has $n_1 = 3$ product lines in industry $i = 1$, $n_3 = 2$ lines in $i = 3$, and $n_4 = 1$ line in $i = 4$; thus, firm f has $n = 6$ product lines in total. This firm does not currently operate in industry $i = 2$.

FIGURE 1: PAYOFF-RELEVANT STATE $\mathbf{q}$

$$\mathbf{q}(t) \equiv \left\{ \begin{array}{ccc} \underline{i=1} & \underline{i=2} & \underline{i=3} & \underline{i=4} \\ q_{1,j_1}(t) & & q_{3,j_1}(t) & q_{4,j_1}(t) \\ q_{1,j_2}(t) & & q_{3,j_2}(t) & \\ q_{1,j_3}(t) & & & \end{array} \right\}.$$

A firm's portfolio of products will expand through successful innovation. Likewise, it will lose product lines when other firms or potential entrants successfully innovate on one of its product lines (thus surpassing it). These innovations will be the source of economic growth in this economy. The next subsection will describe the details of the innovation technology.

2.1.2 Innovation and Technological Progress

In this economy, there are two types of innovations (basic and applied) and two different groups of agents (private and public sectors) generating productivity growth. As Nelson (1959) and Aghion and Howitt (1996) describe it, fundamental advances in technological knowledge come through basic innovation and open up windows of opportunity for future research. Applied innovation builds on these existing basic innovations, thus realizing these opportunities.

Research by Private Firms For the innovation production function, we will follow the literature,⁹ but extend it to both (a)pplied and (b)asic research. More specifically, firms undertake s -type research ($s \in \{a, b\}$) operating with the following convex cost function:

$$h_s(s_i) = \zeta_s s_i^{\nu_s} \quad (5)$$

in terms of researchers per product line, where $s_i \in \{a_i, b_i\}$ is the s -type innovation Poisson flow per product line, ζ_s is a scale parameter, and $\nu_s > 1$ is the cost elasticity.¹⁰ Note that this production

⁸A multi-set is a generalization of a set that can contain more than one instance of the same member. For example, given $j \neq j'$, a multiset $\mathbf{q}_{if}$ can contain $q_{if}(j)$ and $q_{if}(j')$ regardless of whether $q_{if}(j) = q_{if}(j')$.

⁹See, for instance, Klette and Kortum (2004), Lentz and Mortensen (2008), Acemoglu, Akcigit, Hanley, and Kerr (2016), and Acemoglu, Akcigit, Alp, Bloom, and Kerr (2018).

¹⁰This per-product cost function can be derived from a firm-level production function that combines (in a Cobb-Douglas manner) the number of product lines n_i and total research employment H_i to yield total innovation output S_i , as in

$$S_i = \zeta_s^{-\frac{1}{\nu_s}} n_i^{1-\frac{1}{\nu_s}} H_{s,i}^{\frac{1}{\nu_s}}.$$

is specified at the product line level, and a firm's total innovative capacity scales additively across product lines.

Both applied and basic research are *directed* toward particular industries but *undirected* within those industries. In other words, once a firm chooses s_i , the realization of innovations will take place on a random product within industry i . Once this occurs, the newly innovating firm will replace the incumbent firm as the monopolist.¹¹

In our data, some firms do not invest in basic research. To capture this fact, we generalize the basic research technology by introducing a fixed cost of doing basic research. At each instant, a firm with n product lines draws a fixed labor cost of doing basic research $n\phi \geq 0$, where ϕ is distributed according to the distribution $\mathcal{B}(\cdot)$ having mean $\bar{\phi}$.

2.1.3 Spillovers of Basic Research

Let $q_{ij}(t)$ be the highest productivity technology for producing j in industry i . When a firm that operates in industry i produces a basic innovation in industry i and product line j , the same firm uses this basic knowledge for production and patents this new high-value technology. As a result, the firm improves $q_{ij}(t)$ by $\eta\bar{q}_i(t)$

$$q_{ij}(t + \Delta t) = q_{ij}(t) + \eta\bar{q}_i(t), \quad (6)$$

where $\eta > 0$ is the step size drawn from exponential distributions with mean $\bar{\eta}$, and $\bar{q}_i$ is the productivity index defined in equation (4). Notice that this additive formulation features mean reversion in terms of proportional step sizes and, therefore, generates an invariant firm-size distribution over the normalized quality ($q_{ij}/\bar{q}_i$). When the firm produces this new innovation, it adds this product line with the productivity improvement to its portfolio $\mathbf{q}(t + \Delta t) = \mathbf{q}(t) \cup \{q_{ij}(t + \Delta t)\}$, which generates per-period profit of $\pi(q_{ij}(t + \Delta t))$. For instance, when the firm in Figure 1 is successful with its basic research investment in industry $i = 1$, its number of products goes up from $n_1 = 3$ to $n_1 = 4$, as illustrated in Figure 2:

FIGURE 2: IMPACT OF BASIC INNOVATION IN INDUSTRY $i = 1$

$$\mathbf{q}(t) \equiv \left\{ \begin{array}{cccc} \underline{i=1} & \underline{i=2} & \underline{i=3} & \underline{i=4} \\ q_{1,j_1}(t) & & q_{3,j_1}(t) & q_{4,j_1}(t) \\ q_{1,j_2}(t) & & q_{3,j_2}(t) & \\ q_{1,j_3}(t) & & & \\ q_{1,j_4}(t) + \eta\bar{q}_1(t) & & & \end{array} \right\}.$$

Basic research features two potential spillovers: (a) *cross-industry spillover* and (b) *within-industry spillover*. These spillovers lie at the heart of our analysis; therefore, we will now discuss each in more detail.

¹¹Note that the implicit assumption here is that patents grant a monopoly to the leading firm but provide no blocking power against new innovators.

Cross-Industry Spillover from Basic Research The characteristic feature of basic research that we wish to capture is that it often has applications in many industries other than the one for which it was originally intended. Therefore, we will assume that when a basic innovation occurs, it applies with probability one to the target industry i (as illustrated above in Figure 2) and generates a random number of “additional” spillovers in other (or the same) industries. We denote the expected number of such additional spillovers with $p > 0$. Thus, p is our measure of the intensity of cross-industry spillovers. Note that not every spillover can be utilized by the firm. When a firm generates some basic knowledge, it can turn this into an immediate application only in the industry in which it has working knowledge. This structure captures the hypothesis put forth by Nelson (1959).

Going back to our earlier example in Figures 1 and 2, we can now illustrate how cross-industry spillovers operate. Thanks to its success, the firm has already produced a new basic innovation in industry $i = 1$. Assume, now, that this basic innovation has “additional” spillovers in industries 1, 2, and 3 but not in $i = 4$. Since firm f already operates in industries 1, 3, and 4, it can turn this new finding into patentable products in industries 1 and 3, as illustrated in Figure 3.

FIGURE 3: CROSS-INDUSTRY SPILLOVER INTO INDUSTRY $i = 3$

$$\mathbf{q}(t) \equiv \left\{ \begin{array}{cccc} \underline{i=1} & \underline{i=2} & \underline{i=3} & \underline{i=4} \\ q_{1,j_1}(t) & & q_{3,j_1}(t) & q_{4,j_1}(t) \\ q_{1,j_2}(t) & & q_{3,j_2}(t) & \\ q_{1,j_3}(t) & & q_{3,j_3}(t) + \eta \bar{q}_3(t) & \\ q_{1,j_4}(t) + \eta \bar{q}_1(t) & & & \\ q_{1,j_5}(t) + \eta \bar{q}_1(t) & & & \end{array} \right\}.$$

We model the randomly realized number of spillovers generated with a geometric distribution. That is, the probability of receiving $n \geq 0$ spillovers (in addition to the original innovation) is given by

$$F^n = (1 - s)s^n \quad \text{where} \quad s = \frac{p}{1 + p}. \quad (7)$$

Note that the number of spillovers has expectation p by construction. Recall that m denotes the number of industries in which a firm has working knowledge. As we show in Appendix D, this leads to the following distribution of utilized spillovers (k) for a firm with working knowledge in m industries:

$$F_m^k = (1 - s_m)s_m^k \quad \text{where} \quad s_m = \frac{sm}{sm + (1 - s)M} \leq s. \quad (8)$$

Consequently, the expected number of utilized spillovers for the firm is

$$\text{Utilized_Spillovers}(m) = p \times \frac{m}{M}.$$

This highlights the well-known *appropriability problem* of basic research. As Nelson (1959) puts it, firms that have “*fingers in many pies*” – i.e., m in our model – have a higher probability of using the results of basic research. Hence, a broad technological base increases the probability of benefiting from successful basic research.

Within-Industry Spillover from Basic Research *Applied research* makes use of the *within-industry* spillover from basic research and builds on the existing latest basic technological knowledge in a product line. The productivity of each applied innovation is a function of the depreciation of the latest basic technology. If the latest basic knowledge in j is undepreciated (i.e., still hot), a successful applied innovation will benefit from it and improve the latest productivity $q_{ij}(t)$ of that product line by $\eta \bar{q}_i(t)$, as in expression (6). If the latest basic technology of the product line is depreciated (i.e., cold), a successful applied innovation will improve the latest productivity only by an amount proportional to λ , so that

$$q_{ij}(t + \Delta t) = \begin{cases} q_{ij}(t) + \eta \bar{q}_i(t) & \text{if } j \text{ is hot} \\ q_{ij}(t) + \lambda \bar{q}_i(t) & \text{if } j \text{ is cold} \end{cases} \quad (9)$$

Note that the step sizes η and λ are drawn each time from an exponential distributions with mean $\bar{\eta}$ and $\bar{\lambda}$.

We assume that a new basic technology depreciates (innovations run into diminishing returns) at a Poisson rate $\zeta > 0$. On the other hand, a new basic innovation reactivates the product line until the next time that it cools down again. If the product line was already hot, there is no additional effect. Let us denote the arrival rate of basic innovations to product lines by τ_b . Then, during a small time interval Δt , each product line j will be subject to the transition rates denoted in Table 1:

TABLE 1: TRANSITION MATRIX FOR WITHIN-INDUSTRY SPILLOVERS

	hot	cold
hot	$1 - \zeta \Delta t$	$\zeta \Delta t$
cold	$\tau_b \Delta t$	$1 - \tau_b \Delta t$

Public Basic Research In our model, the academic sector will be the other source of basic knowledge creation. One of the main tasks of public research labs in an economy is to produce the necessary basic scientific knowledge that will be part of the engine for subsequent applied innovations and growth. We assume that the public research sector consists of a measure U of research labs *per industry*. Each lab receives the same transfer $\bar{R}$ from the government to finance its research, which results in an overall funding level of $R = \bar{R} \times U \times M$.

We assume that each public research lab generates a flow rate of u by hiring $h_b(u)$ researchers with the same basic research technology as a one-product firm in (5), so that $h_b(u) = \zeta_b u^{\nu_b}$.¹² Additionally, as with private basic research, they face a per-lab fixed cost $\bar{\phi}$ of operation. This specification implies that the government can affect the basic knowledge pool in the economy through the amount of funds R allocated to the academic sector. The flow rate of basic innovation from the academic sector will

¹²In reality, public research labs may have a different research technology than private labs. However, obtaining data on both the inputs and outputs of individual public labs is difficult. The separate estimation of public and private innovation production functions is left for future research.

satisfy

$$\bar{R} = w[h_b(u) + \bar{\phi}], \quad (10)$$

where u is the academic basic innovation flow *per lab*. In this economy, R is a policy lever controlled by the policymaker. As with private firms, each basic innovation has the same within- and cross-industry spillover structure. Note that the equilibrium fraction of hot product lines α will be determined by the aggregate rates of public (u) and private (b_m) basic research, as well as by the cooldown rate (ζ). The key difference between private and public basic research is the "Ivory Tower" nature of the latter, meaning that these innovations by public labs will turn into output only upon a subsequent private applied innovation.¹³

Research and Innovation Policy In the baseline model, we consider proportional subsidies to research spending. We allow for there to be differential subsidy rates ψ_a and ψ_b for applied and basic, respectively. In the policy analysis, we consider optimal policies for the case where there is a uniform subsidy ($\psi_a = \psi_b$) and where the subsidy rates can differ ($\psi_a \neq \psi_b$).

2.1.4 Entry, Exit, and Industry Dynamics

Entry There is a mass E of outside entrants per industry. The research technology for a single outside entrant is assumed to be the same as that of applied innovation for a firm with a single product line. Thus, if an outside entrant wishes to produce a flow probability a_e of entry, they must hire $h_a(a_e) = \xi_a a_e^{v_a}$ researchers.

Exit In our model, there will be both endogenous and exogenous channels for firm exit. First, a firm that loses all of its product lines to other competitors will have a value of zero and, thus, will exit. Second, each firm has an exogenous death rate $\kappa > 0$. When this occurs, the firm sells all of its product lines to random firms at a "fire sale" price $\mathcal{P}$.¹⁴ On the flip side, firms will receive a buyout option with a probability that is proportional to their number of products.

Expansion into New Industries The economy features $E \times a_e$ flow of entry at any instant. We will assume that a ζ fraction of new entrants will meet a randomly selected incumbent firm. Thus, an incumbent will have a flow rate of incoming buyout offers

$$x \equiv \zeta E a_e / F,$$

where F is the equilibrium measure of firms. If $\bar{n}$ denotes the average number of product lines per firm, then $F = 1/\bar{n}$. Clearly, this new company will be from a new industry with probability

¹³It is important to note that we assume that innovation done by public labs is turned into consumer products only upon subsequent innovation by private firms. The lag between the creation of publicly funded innovations and actual goods production is empirically shown in a large literature (e.g., [Rosenberg and Nelson \(1994\)](#) and [Mowery, Nelson, Sampat, and Ziedonis \(2004\)](#), among several others). This important issue is generally overlooked in the theoretical growth literature. Inclusion of this feature generates some new and interesting dynamics, such as the importance of involvement of the private sector in basic research.

¹⁴The exact value of this price will not play any role in the equilibrium determination.

$(1 - m/M)$ or from an industry that already exists in the incumbent's portfolio with probability m/M . Our goal is to keep the M&A margin as tractable as possible, and we will achieve this by assuming that the M&A price that the incumbent firm has to pay is equal to the full surplus of the new merger.¹⁵

Labor Market Labor is split between production (L_p) and research labor. Research labor can be further subdivided into that devoted to private basic research (L_b), public basic research (L_u), private applied research (L_a), and firm entry (L_e). There is a unit mass of workers per industry, meaning the total labor supply consists of M workers. The labor market clearing condition is given by

$$M = L_p + L_b + L_a + L_e + L_u.$$

Household Problem Finally, we close the model by describing the household problem that determines the equilibrium interest rate in this model. The household consumes the final good and maximizes the following lifetime utility

$$W_0 = \int_0^\infty \exp(-\delta t) \frac{C(t)^{1-\gamma} - 1}{1-\gamma} dt, \quad (11)$$

where $C(t)$ is consumption at time t ; γ is the constant relative risk-aversion parameter; and δ is the discount rate. The household owns all the firms in the economy, which generates a risk-free flow return of r in aggregate. The household also supplies labor in the economy, through which it earns wage rate $w(t)$. Finally, the household pays a lump-sum tax $T(t) \geq 0$ every instant. Thus, the household's intertemporal maximization is simply to maximize (11) subject to the following budget constraint:

$$C(t) + \dot{A}(t) \leq r(t) A(t) + Mw(t) - T(t),$$

where $A(t)$ is the asset holdings of the household.

2.2 Equilibrium

In this section, we characterize the dynamic equilibrium of our model. Our focus is on a symmetric balanced-growth-path (SBGP) equilibrium in which all industries start with the same initial conditions at time $t = 0$, and all aggregate variables grow at the same endogenous rate g .

In this model, three variables affect the payoff of the firm: the number of product lines n ; the number of industries m , and the relative productivity

$$\hat{q}_{ij} \equiv q_{ij} / \bar{q}_i \quad (12)$$

of its product lines, which is the absolute productivity in line j normalized by the productivity index $\bar{q}_i$ in industry i . Note that this implies that successful innovation leads to a constant additive increment

¹⁵The resulting invariant joint distribution $\Gamma_{m,n}$ over multi-industry presence m and firm product count n is described in Appendix D

in $\hat{q}$ space, that is, $\hat{q} \rightarrow \hat{q} + s$ where $s \in \{\eta, \lambda\}$ depends on the type of innovation. Thus, each incumbent firm is characterized by its state $k \equiv (\hat{\mathbf{q}}, n, m)$.

Of particular interest is the distribution of firms $\Gamma_{n,m}$, which denotes the mass of firms operating in m industries and having n product lines in total. This will sum not to one but to the total number of firms, which is endogenous. We also denote the mass of product lines owned by firms in m industries by μ_m , which can be computed from the joint distribution using $\mu_m \equiv \sum_{n=1}^{\infty} n \cdot \Gamma_{m,n}$. This will in fact sum to one, as the total mass of product lines in each industry is also one.

Given a government policy sequence $[T(t)]_{t=0}^{\infty}$, an SBGP equilibrium is composed of a sequence of intermediate-good quantities; prices; the basic and applied innovation rates of private firms and entrants; the wage rate and interest rate; the joint distribution of multi-industry presence and product count; hot and cold product line productivity distributions $(\mathcal{F}_H(t), \mathcal{F}_C(t))$; and the fraction of hot product lines – i.e., $[y_k(t), p_k(t), b_k(t), a_k(t), a_e(t), w(t), r(t), \Gamma_{m,n}(t), \mathcal{F}_H(t), \mathcal{F}_C(t), \alpha(t)]_{t=0}^{\infty}$ – such that all firms choose quantity and price to maximize their profits; incumbent and entrant firms invest in research to maximize their firm value; the labor market clears; the household maximizes its discounted sum of future utilities; and the distributions satisfy the relevant flow equations.

Solution of the Model In this setting, the intermediate producer with the state-of-the-art technology $\hat{q}$ operates as a monopolist. Thus, each new innovation has an element of creative destruction, wherein the new innovator replaces the old incumbent using superior technology. In order to avoid the possibility of limit-pricing, we follow [Acemoglu, Akcigit, Alp, Bloom, and Kerr \(2018\)](#) and [Akcigit and Kerr \(2018\)](#), and assume that the current incumbent and any former incumbents in the same line (with lower quality than the current incumbent) enter a two-stage price-bidding game. In the first stage, each firm pays a fee of $\epsilon > 0$, which is arbitrarily close to 0. In the second stage, all firms that paid the fee announce their prices. Due to Bertrand competition in the second stage, only the most productive firm will be able to make any sales and profits, and thus, only this firm will pay the cost ϵ , enter the pricing game, and operate with monopoly pricing.

Standard profit maximization delivers familiar equilibrium price and quantities (interested readers are referred to Appendix Section D for the detailed derivations and results). A monopolist's quantity is increasing and price is decreasing in the relative productivity $\hat{q}$ of the product line. The equilibrium profits of the monopolist are increasing in its relative productivity $\hat{q}$ and the average market size Z/M :

$$\pi(\hat{q}) = \frac{\hat{q}^{\epsilon-1}}{\epsilon} \frac{Z}{M}. \quad (13)$$

From this expression, for the flow profits at the product line level, we can then construct a value function describing the overall value of a firm with a portfolio $\hat{\mathbf{q}}$ of product lines operating in m industries. This rather ungainly object is presented in full in Appendix D. However, the intuition for its construction is fairly straightforward. There is a flow value associated with profits net of various types of R&D spending. On top of this, there are a variety of events that can happen to the firm, such as gaining a new product line on successful research, expanding into a new industry, or

losing a product line to a competitor who has innovated. Proposition 1 summarizes our main result concerning firm value.

Proposition 1. *Let the value of a firm with a productivity portfolio $\hat{\mathbf{q}}$ in m industries be denoted by $\mathcal{V}(\hat{\mathbf{q}}, m)$. This value is equal to*

$$\mathcal{V}(\hat{\mathbf{q}}, m) = \frac{Z}{M} \left[\sum_{\hat{q} \in \hat{\mathbf{q}}} V(\hat{q}) + nV_m \right],$$

where

$$V(\hat{q}) = \frac{\hat{q}^{\varepsilon-1}}{\varepsilon [r + \tau + \kappa + g(\varepsilon - 2)]},$$

and

$$(r - g) V_m = \max_{a,b} \left\{ \begin{array}{l} -\tilde{w} \left[(1 - \psi_a) h_a(a) + (1 - \psi_b) [h_b(b) + \mathbf{1}_{(b>0)} \phi] \right] \\ + a \left[\alpha V^H + (1 - \alpha) V^C + V_m \right] + b \left(1 + p \frac{m}{M} \right) [V^H + V_m] \\ + x \left(1 - \frac{m}{M} \right) [V_{m+1} - V_m] - \tau V_m + \kappa \mathbb{E}_{\hat{q}} V(\hat{q}_t) \end{array} \right\}. \quad (14)$$

The analogous production values are defined as $V^H \equiv \mathbb{E}_{\hat{q}, \eta}^H V(\hat{q} + \eta)$ and $V^C \equiv \mathbb{E}_{\hat{q}, \lambda}^C V(\hat{q} + \lambda)$.

Proof. See Appendix D ■

This important result has a number of implications. First, the value of a firm has a tractable additive form across product lines. Moreover, the firm value has two major components: the first component is the production value $V(\hat{q})$, which simply computes the sum of the future discounted profits where the effective discount rate takes into account the rate of creative destruction τ , the exogenous destruction rate κ , and the obsolescence of the relative productivity $\hat{q}$ due to the growth of $\bar{q}$. The second component is the R&D option value V_m , which is a direct function of the multi-industry presence due to the associated internalization of spillovers. Finally, because of the stochastic nature of step sizes, the expectations now integrate over the productivity (which is type specific) and step size.

Because the value function is additive across product lines, so then are innovation rates. Now we can express the first-order condition for basic innovation at the product line level with

$$b_m = \left[\frac{(1 + p \frac{m}{M}) (V^H + V_m)}{(1 - \psi_b) v_b \tilde{\zeta}_b \tilde{w}} \right]^{\frac{1}{v_b - 1}}$$

The most important result here is the fact that the cross-industry spillover is stronger when a firm is operating more industries; thus, we would expect that basic research investment is increasing in the multi-industry presence of the firm. The strength of this positive relationship will be governed mainly by the probability of the cross-industry spillover parameter p . Because of the fixed cost to undertaking basic research ϕ , some labs will decide not to operate if their draw is too high. Let this threshold value be denoted ϕ_m . Furthermore denote the resulting fraction of labs that operate be denoted $P_m = \mathbb{P}[\phi < \phi_m]$.

Both private firms and public research labs are generating basic research in this economy. It is useful to break down total basic research into its embodied and disembodied components. The distinction is based on whether the basic knowledge is immediately turned into a consumer product

(embodied) or simply added to the stock of knowledge available for future innovators (disembodied). We obtain the following aggregates:

$$\begin{aligned}
 \text{Embodied: } \tau_b^e &\equiv \sum_{m=1}^M \mu_m P_m (1 + \rho_m) b_m \\
 \text{Disembodied: } \tau_b^d &\equiv \sum_{m=1}^M \mu_m P_m (p - \rho_m) b_m + (1 + p)u \\
 \text{Total: } \tau_b &\equiv \tau_b^e + \tau_b^d,
 \end{aligned} \tag{15}$$

Recall that μ_m is the mass of product lines owned by firms operating in m industries μ_m . Then, τ_b^e and τ_b^d correspond, respectively, to the embodied and disembodied components of basic research. Note that the disembodied component includes both private spillovers that are unused and the results of public basic innovation. Finally, τ_b is simply the overall flow of basic innovation, including all spillovers.

Using this aggregate rate and the cooldown rate ζ , we can express the steady-state flow equation: the number of product lines that become hot must be equal to the number of product lines that cool down. In other words, we must have $\alpha\zeta = (1 - \alpha)\tau_b$. As a result, the steady-state fraction of hot product lines is

$$\alpha = \frac{\tau_b}{\zeta + \tau_b}. \tag{16}$$

The share of hot product lines – those having basic knowledge that can be turned into better consumer products (α) – is increasing in the amount of basic research flow. This expression highlights the role of public policy in affecting the knowledge stock. The more money that is allocated to public basic research, the higher the basic research flow will be from public research labs (u), which will then increase the fraction of hot product lines through τ_b , as in (15) and (16).

Firms invest in applied research according to

$$a = \left[\frac{\alpha V^H + (1 - \alpha) V^C}{(1 - \psi_a) \nu_a \zeta_a \bar{w}} \right]^{\frac{1}{\nu_a - 1}}.$$

The crucial observation here is the complementarity between basic and applied research. In equilibrium, $V^H > V^C$, since hot product lines are associated with a larger step size η . Hence, if there are more hot product lines (a higher α), each firm increases its investment in applied research. This renders basic and applied research investments complementary. In particular, holding the wage rate fixed, higher public basic research investment encourages firms to invest more in applied research through higher within-industry spillovers.

Let us denote the aggregate rate of applied innovation by τ_a such that

$$\tau_a = \sum_{m=1}^M \mu_m a_m + E a_e. \tag{17}$$

Recall that τ_b^e denotes the arrival rate of embodied basic research, as defined in (15). Now we can denote the aggregate rate of *creative destruction* (the rate at which firms lose product lines to other

firms) by τ :

$$\tau \equiv \tau_a + \tau_b^e. \quad (18)$$

Creative destruction is determined by the rate at which incumbents produce basic innovations that can be embodied into production immediately (τ_b^e), and by the rate at which incumbents and entrants produce applied innovations (τ_a). Now we are ready to state the following lemma:

Lemma 1. *Let $\mathcal{F}_H(\cdot)$ and $\mathcal{F}_C(\cdot)$ be the aggregate product cumulative measures by type (hot or cold). The flow equations for these objects are, respectively,*

$$\begin{aligned} \dot{\mathcal{F}}_H(\hat{q}) &= -\tau [\mathcal{F}_H(\hat{q}) - \mathcal{F}_H(\hat{q} - \eta)] + \tau_b^e \mathcal{F}_C(\hat{q} - \eta) - \zeta \mathcal{F}_H(\hat{q}) + \tau_b^d \mathcal{F}_C(\hat{q}) + g\hat{q}[\partial \mathcal{F}_H(\hat{q}) / \partial \hat{q}] \\ \dot{\mathcal{F}}_C(\hat{q}) &= -\tau_a [\mathcal{F}_C(\hat{q}) - \mathcal{F}_C(\hat{q} - \lambda)] - \tau_b \mathcal{F}_C(\hat{q}) + \zeta \mathcal{F}_H(\hat{q}) + g\hat{q}[\partial \mathcal{F}_C(\hat{q}) / \partial \hat{q}]. \end{aligned}$$

Proof. See Appendix D. ■

The labor market clearing condition can now be expressed in terms of the above endogenous variables. One additional relationship we will exploit is that between the mass of labor devoted to production and the normalized wage rate. This can be derived from the goods production specification (see Section D in the Appendix for its detailed derivation)

$$L_p = \frac{Z}{w} \left(\frac{\varepsilon - 1}{\varepsilon} \right)$$

Using this and the symmetric nature of the equilibrium, we express the labor market clearing condition as an average over industries:

$$1 = \frac{1}{\bar{w}} \left(\frac{1 - \varepsilon}{\varepsilon} \right) + \sum_m \mu_m P_m [h_b(b_m) + \bar{\phi}_m] + U[h_b(u) + \bar{\phi}] + h_a(a) + E h_a(a_e) \quad (19)$$

where $\bar{\phi}_m = \mathbb{E}[\phi | \phi < \phi_m]$. This expression equates the labor supply per industry (= 1 since the total labor supply is M) to labor demand for production workers; private basic researchers, which is a function of the multi-industry presence of the firms; public basic researchers, which is determined by public policy; incumbent applied researchers; and entrant basic researchers.

Finally, as derived in the Appendix, by imposing aggregate consistency with the goods production functions (2) and (1), we find that the aggregate output is

$$Z = \bar{q} L_p / M. \quad (20)$$

This expression simply says that the aggregate output is equal to the product of the number of workers employed for production and the aggregate productivity index of the economy. In an SBGP equilibrium, the labor allocated for production is constant. Therefore, the growth rate of aggregate output (and also output per worker) will be equal to the growth rate of the productivity index $\bar{q}$. The following proposition provides the exact growth rate of the productivity index.

Proposition 2. *In an SBGP, the growth rate of the productivity index is*

$$g = \frac{\tau_a \left[\alpha \mathbb{E}_{\hat{q}}^H (\hat{q} + \eta)^{\varepsilon-1} + (1 - \alpha) \mathbb{E}_{\hat{q}}^C (\hat{q} + \lambda)^{\varepsilon-1} - 1 \right] + \tau_b^e \left[\mathbb{E}_{\hat{q}} (\hat{q} + \eta)^{\varepsilon-1} - 1 \right]}{\varepsilon - 1}. \quad (21)$$

Proof. See Appendix D ■

This growth expression shows that the engines of economic progress include both applied and basic innovation. More important, the basic knowledge stock in the economy, represented by α , makes each applied innovation more valuable and contributes more to growth (since $\eta > \lambda$). This expression shows how public funding can contribute to growth through its indirect impact on private research.

2.3 Welfare Properties

Finally, we close this section by describing the SBGP equilibrium welfare. In an SBGP equilibrium that has an initial consumption C_0 and a growth rate of g , welfare is computed as

$$W(C_0, g)^{SBGP} = \int_0^\infty \exp(-\delta t) \frac{(C_0 e^{gt})^{1-\gamma} - 1}{1-\gamma} dt = \frac{1}{1-\gamma} \left(\frac{C_0^{1-\gamma}}{\delta - (1-\gamma)g} - \frac{1}{\delta} \right).$$

We will report our results in consumption-equivalent terms. In particular, when two different public policies T_1 and T_2 generate different SBGP equilibrium welfare values as $W(C_0^{T_1}, g^{T_1})$ and $W(C_0^{T_2}, g^{T_2})$, we will report β such that

$$W(\beta C_0^{T_1}, g^{T_1}) = W(C_0^{T_2}, g^{T_2}).$$

In other words, β constitutes the compensating differential in initial consumption that equalizes the welfare of the two proposed policy environments. Therefore, it provides an intuitive measure for evaluating policy tools.

To sum up, let us briefly discuss the sources of inefficiency and what policy can achieve in this model. First, as in standard quality ladder models, there are intertemporal spillovers within each product line. Second, firms simply enjoy the expected duration of monopoly power due to the competition channel of creative destruction. As a result, the private value of innovation differs from the social value of innovation. It is also worth highlighting that, in this model, there could be either over- or underinvestment in R&D. In addition to the standard channels, our model features additional spillovers due to basic research, both within and across industries. Finally, there are additional static distortions due to monopoly power. However, since we are interested primarily in the dynamic inefficiencies associated with innovation and basic research, we will consider the case of a social planner who is still subject to monopoly distortions on the production side.

In the full dynamic framework, all of these inefficiencies will generate room for innovation policy, and our estimated model will govern whether there is over- or underinvestment in the various types of research expenditures in the decentralized equilibrium. It will also provide a framework within which we can evaluate the effects of these innovation policies.

2.4 Social Planner

For the social planner's problem, we choose innovation rates to maximize steady state welfare, while taking static monopoly distortions as given. Furthermore, we assume that all spillovers from both the public and private research sectors are immediately internalized. This allows us to make a number

of simplifications. First, both the labor cost and growth impact of applied innovation, be it by incumbents or entrants, become the same, so we need only choose a common rate of applied innovation a for a mass $1 + E$ of applied research labs. Similarly, the same is true for private and academic basic research, and so we need only choose a common rate of basic innovation b . There is still the extensive margin decision for private basic research labs, which we characterize with a threshold value $\hat{\phi}$ for the random fixed cost.

As seen in equation (20), initial output (and hence consumption) is simply a function of the production labor allocation, which is equal to one minus the research labor allocation. Thus, we can perform the following constrained maximization over the logarithm of welfare

$$\begin{aligned}
 \max_{a,b,\hat{\phi}} \quad & (1 - \gamma) \log(1 - R) - \log(\delta - (1 - \gamma)g) \\
 \text{s.t.} \quad & P = \mathbb{P}[\phi < \hat{\phi}] \\
 & \tau_a = (1 + E)a \\
 & \tau_b = (P + U)(1 + p)b \\
 & R = (1 + E)h_a(a) + (P + U)h_b(b) + P\mathbb{E}[\phi | \phi < \hat{\phi}] + U\bar{\phi} \\
 & g = g(\tau_a, \tau_b) \quad (\text{see eq. 21})
 \end{aligned} \tag{22}$$

Before moving on to the quantitative analysis, it is useful to provide a discussion of the key assumptions of our model and their implications on the results. This also allows us to better convey the theoretical structure underpinning our quantitative estimates and the associated policy results.

2.5 Modelling Choices and Assumptions

As in other endogenous growth models, innovations in our model occur thanks to the monopoly rents that are protected by patents.¹⁶ Since patent protections are in place, competitors cannot simply copy the frontier technology, and they need to come up with a technology that is superior to the incumbent's technology. This business stealing effect introduces a source of overinvestment in R&D. At the same time, firms are building on the shoulders of giants (i.e., past innovations before time t make the current innovation step size increase by $\bar{q}_i(t)$), introducing a source of underinvestment in R&D. Hence, our choice of Schumpeterian models provides the possibility that the decentralized economy could feature over- or underinvestment in R&D.

The reader should note that we have full business stealing, in the sense that entry completely pushes out the previous incumbent in that product line. In an alternative specification, one could imagine some type of patent sale or licensing arrangement. Such an arrangement would induce the need for payments between incumbents and new entrants. Consequently incumbents would potentially have blocking power and receive a stream of royalties. This might affect the model's efficiency properties as they relate to within industry spillovers, but would likely not change the

¹⁶To see an explicit treatment of patent policy in this class of models, see [Acemoglu and Akcigit \(2012\)](#), among others.

inefficiencies arising from uninternalized cross-industry spillovers. In our model, we abstract from such possibilities mostly because, to the best of our knowledge, there is no systematic data on such licensing transactions that could discipline this margin.

Another important feature of the model is that creative destruction happens because rival firms are innovating with a technology that makes the previous technology (and its patent) obsolete. Our model therefore ignores the possibility that firms may improve their own products through internal innovation, without stealing someone else's product through external innovation. Introducing internal innovations into this model would have an ambiguous impact on innovation incentives. On the one hand, firms would internalize the intertemporal spillovers that the internal innovation generates, and increase their innovation efforts. On the other hand, this mechanism would reduce the business stealing margin, and therefore reduce investment into research. However, as in the previous discussion, the inefficiency properties associated with the cross-industry spillover should remain the same. Empirically, there is some disagreement in the literature about the relative importance of internal versus external innovations.¹⁷ Hence, while we appreciate the fact that this additional distinction could be interesting to incorporate, we prefer to leave this as an extension for future research.

A key distinction introduced by our model is the split between basic and applied research. Importantly, the model structure has enough flexibility to allow for quantitatively different or similar spillovers within and across industries, depending on the estimated step sizes (η, λ) . In addition, since the government is also investing in basic research through public research labs, it could be that, depending on how much money is allocated to them, the decentralized economy over- or under-invests in basic research relative to applied research. Therefore, the efficiency properties become a quantitative question, and our model has the ability to stimulate meaningful policy discussions in this domain.

Finally, in addition to distinguishing between basic and applied research, we also consider whether research is undertaken by the private or the public sector. Universities are a major source of basic research in our model, and an important simplifying assumption is that their only goal is to improve productivity. This could be a strong assumption since universities and policymakers also care about other educational aspects of knowledge creation and knowledge diffusion. While it is difficult to precisely quantify the weight attached by policymakers to each objective, such additional benefits would lead us to under-estimate the optimal allocation of funds to the public research sector. In our model, such additional objectives could be introduced by changing the objective function of the social planner.

3 Data, Measurement, and Empirical Evidence

In this section, we present and discuss the empirical patterns that provide the basis for the structural estimation in Section 4. First, we describe the data sources that allow us to document patterns of research investment and patenting decisions. Second, we show how the research investment patterns

¹⁷For instance see [Garcia-Macia, Hsieh, and Klenow \(2019\)](#) and [Akcigit and Kerr \(2018\)](#).

of diversified firms allow us to identify cross-industry spillovers of research. We then use patent citation data across public and private inventors to quantify step sizes in basic and applied research, as well as the associated cooldown rate.

3.1 Data and Variable Construction

To test and estimate our model, we rely on a unique combination of datasets on French innovative firms between 2000 to 2006. The final sample is composed of 13,708 firm-year observations, for which we provide further details regarding data organization and descriptive statistics in Appendix A.

R&D Information Information about research investment comes from the annual R&D Survey conducted by the French Ministry of Research. The survey is conducted in annual waves of cross-sectional data, and covers a representative sample of French firms of more than 20 employees investing in R&D. The questionnaire includes extensive information about the financing of R&D. It not only breaks down R&D investment according to the source of the funds, but also provides the allocation to different types of research.

The distinction between basic and applied research is based on survey standards defined by the OECD in the Frascati manual. Basic research investment refers to a “systematic study to gain more comprehensive knowledge or understanding of the subject under study without specific applications in mind.” Conversely, applied research is defined as a “systematic study to gain knowledge or understanding to meet a specific, recognized need.”¹⁸

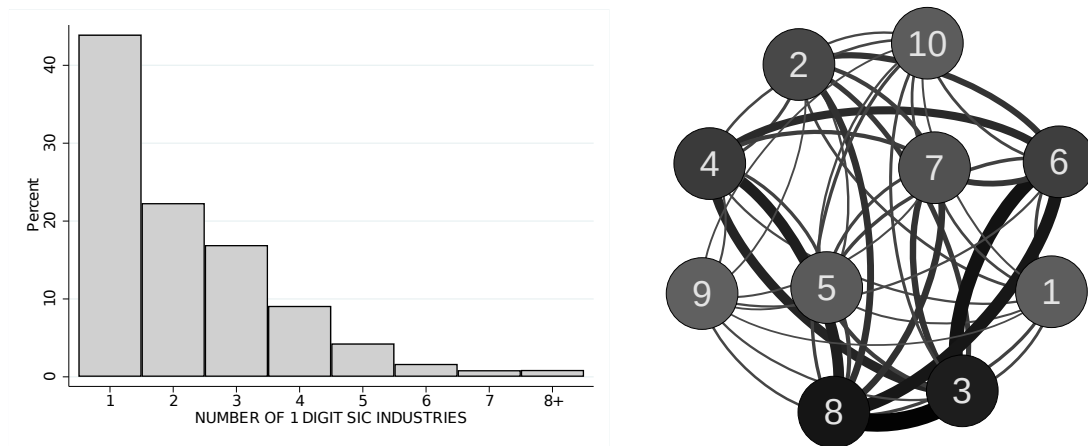
One possible concern is that it is hard to conceptually draw a line between what constitutes basic scientific research and applied scientific research. Indeed, there is a continuous spectrum of scientific activity and “moving from the applied-science end of the spectrum to the basic science end... [.] goals become less clearly defined” (Nelson, 1959). Similarly, Rosenberg (1990) argues that it is challenging to categorize research on the basis of the motives of the person performing it. We address this concern in two ways. First, we note that several seminal papers in the literature (Mansfield, 1980; Mansfield, 1981; Link, 1981; Griliches, 1986) show that the distinction between basic and applied research in this type of data is not only conceptually useful, but also quantitatively important.¹⁹ Second, we also consider the measurement issue from within the context of our model and quantitative results. In particular, we explicitly allow for misclassification of research type in Section 5.5. By doing so, we consider how policymakers could set policies in the presence of various levels of misclassification.

¹⁸Descriptive statistics are provided in Appendix A. One should note that the definition of corporate basic research does not exclude that it may be directed to fields of current or potential interest, as in the examples provided in the introduction. Importantly, our measurement of research expenditures is based on “intramural” activities and allows us to clearly distinguish research developed within the company from research outsourced to universities or to other units in the group.

¹⁹These papers typically evaluate the importance of basic and applied research by relating it to production data and estimating its output elasticity or rate of return from an extended Cobb-Douglas production function. The common conclusion from those papers is that basic research seems to command a significant “premium”—i.e., that its contribution to firm productivity is significantly larger compared to that of applied research.

Multi-Industry Activity & Balance-Sheet Information To measure industry presence, we use information from “Enquete Liaisons Financieres” (LIFI) and “Enquete Annuelle des Entreprises” (EAE). The former allows us to distinguish activities within the same business group, while the latter distinguishes between activities within the same firm and provides us with accounting data. All the data sources are connected through unique firm identifiers allowing us to match them to the R&D data.

FIGURE 4: DISTRIBUTION AND LINKS OF FIRMS ACROSS INDUSTRIES



The figure uses 13,708 firm-year observations from the pooled data for the period 2000-2006. The left panel plots the share of firms as a function of the *number of 1-Digit SIC industries*, which is the number of distinct SIC codes in which a firm is present. The right panel plots the pairwise presence of firms across industries through nodes and lines. Node color is assigned in terms of total number of firms in the industry (lighter to darker). Thickness of lines between industries is assigned in terms of the total number of links between each pair. Activity classification: 1 "Agriculture", 2 "Food and Textile Industries", 3 "Manufacture of chemicals, metals and machinery", 4 "Manufacture of electrical and transport equipment", 5 "Construction and Utilities", 6 "Wholesale and retail trade", 7 "Transport, Communication and Financial Activities", 8 "Professional, scientific and technical activities", 9 "Education and Human Health Activities", 10 "Arts, entertainment and others".

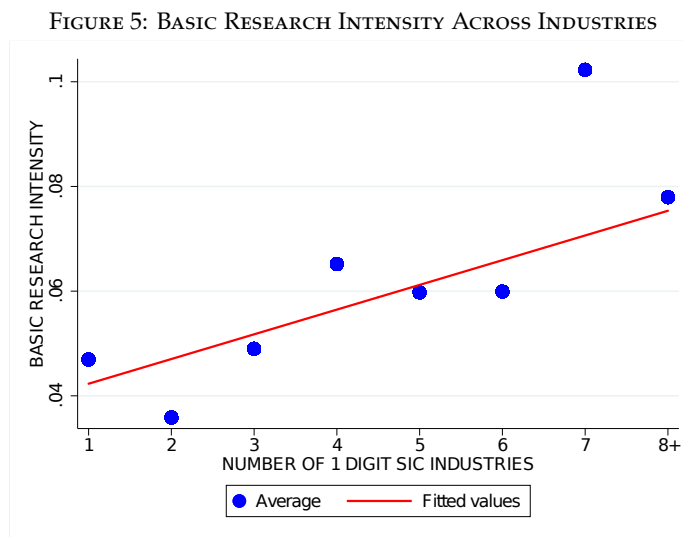
Figure 4 plots the multi-industry presence of firms and the direction of these links. The left panel of Figure 4 plots the share of firms as a function of the number of 1-Digit SIC industries, which is the number of distinct SIC codes in which a firm is present. On average, firms are present in two distinct industries, as defined by 1-digit SIC codes. Although nearly 44% of the firms are operating in only one industry, the remaining firms occupy a large spectrum of industries. Results are very similar when using more disaggregate SIC classifications (up to the 4-digit SIC level) or when changing the definition of an industry link. The right panel of Figure 4 plots the pairwise presence of firms across industries through nodes and lines. Each node color represents the total number of firms in the industry (darker means more firms), while the thickness of the lines between industries represents the total number of links between each industry pair. The figure shows that the most intense industry links flow between manufacturing industries (node 3) and scientific activities (node

8), while pharmaceutical firms (node 9) are less connected to the rest of the innovative firms. One concern is that our measure of multi-industry presence could be driven by the presence of financial units of firms. While some of the links also flow through financial firms included in “Transport, Communication and Financial Activities” (node 7), they do not appear to dominate in any way. In Column (2) of Table 2, Appendix B, we recompute our measures for multi-industry presence and size by excluding financial links and show that our estimate for the cross-industry spillover remains unaffected.

3.2 Investment Across Research Types

We next identify the cross-industry spillovers associated with basic research through the investment choices of multi-industry firms. To do so, we exploit the intuition provided in Nelson (1959): as the range of a firm’s products and industries becomes more diversified, its incentive for investing in basic research relative to applied research should increase due to better appropriability of potential knowledge spillovers.

Figure 5 puts together firms’ basic research investment and their multi-industry presence. It plots the ratio of basic to applied research according to firms’ multi-industry presence. The figure already controls for firm size, measured in terms of log employment, to disentangle it from the multi-industry effect. The figure provides a first visual confirmation that investment in basic research increases as the scope of a firm’s activities expands.



The figure plots *Basic Research Intensity* as a function of *Number of 1-Digit SIC Industries*. For each firm-year observation *Basic Research Intensity* is defined as the ratio of total firm investment in basic research divided by total firm investment in applied research. *Average* is the average basic research intensity for firms conditional on the number of their 1-Digit SIC industries and after partialling out for firm size, while the red line represents a linear fit of the firm-year observations.

Table 2 further strengthens the identification of the cross-industry spillover by adding controls for firm and industry heterogeneity. To account for zeros in the the dependent variable, we estimate a

Tobit model and report marginal effects.

TABLE 2: BASIC RESEARCH AND MULTI-MARKET ACTIVITY - INDUSTRY AND FIRM HETEROGENEITY

	(1)	(2)	(3)	(4)
Log # of Industries	0.032*** (0.006)	0.025*** (0.008)	0.024*** (0.008)	0.024*** (0.008)
Log Employment	0.003** (0.001)	0.002 (0.001)	0.002 (0.001)	-0.000 (0.002)
Year & Organization FE	Yes	Yes	Yes	Yes
Industry FE	No	Yes	Yes	Yes
Areas of Research Shares	No	No	Yes	Yes
Patent Stock & No Patent FE	No	No	No	Yes
N	13708	13708	13707	13708

Notes: Pooled data for the period 2000-2006. Estimates are obtained using Tobit models and relate to the marginal effect of the variables at the sample mean. *Basic Research Intensity* is defined as the ratio of total firm investment in basic research divided by total firm investment in applied research. *Log # of Industries* is the number of distinct SIC codes in which a firm is present. *Log Employment* is the total employment of firms. *Share in Software/Biotech/Materials* is the share of R&D investment in the respective areas of research with share of environment research omitted. *Log Patent Stock* measures the number of patents obtained by firms in the pre-sample period (1993-2000). *Industry FE* denote fixed effects for 1-digit SIC activities. *Year FE* denotes year fixed effects, and *Organization FE* denotes whether the firm operates its activity as a conglomerate or as a business group. See the Appendix for the definition of variables. Robust standard errors clustered at the firm level are in parentheses. One star denotes significance at the 10% level; two stars denote significance at the 5% level; and three stars denote significance at the 1% level.

The baseline specification in column (1) controls for firm size, as measured by log employment. The estimates suggest that a firm's presence in an additional industry is positively correlated with its propensity to invest into more basic research. In terms of magnitude, an additional industry link increases the firm's basic research intensity by three percentage points, on average, corresponding to a 50% increase in the average basic intensity of a single-industry firm.

To better understand whether industry characteristics are the main drivers of our estimated spillover parameter, we proceed in two steps. In Column (2), we first augment the specification relating basic research intensity to multi-industry presence by including a set of industry fixed effects. This specification, therefore, disentangles the spillover effect from industry heterogeneity in research. While the point estimate is slightly lower, the direction and the strength of the correlation remain very similar to the baseline estimates reported in Column (1). A potential concern is that the industry fixed effects might fail to appropriately capture the transversal nature of research. For instance, the multi-industry structure of a firm might be due to its focus on IT research. To address this concern, we exploit information contained in the R&D survey about the amount spent on different areas of research. The survey distinguishes among the following areas: software, biotech, environment, materials, and social sciences. We augment our specification by including controls for areas of

research in column (3), and estimates remain very stable.²⁰

Finally, in column 4, we try to capture firm-specific heterogeneity in innovative capacity. To do so, we follow an insight provided by [Blundell, Griffith, and Reenen \(1995\)](#). Their study estimates the determinants of innovation activity and tries to overcome the fact that “such things as the different appropriability conditions of research efforts and technological opportunities facing firms” are difficult to measure. The authors argue that the permanent firm-specific capacities for successful innovation should be reflected in the pre-sample history of innovative success. Similar to their study, we include the number of patents obtained by firms in the pre-sample period (1993-2000) as a variable that approximates the knowledge stock of the firm at its point of entry into the sample. Again, our estimates remain very similar in economic and statistical terms.

Table 3 tries to empirically address concerns about reverse causality – i.e., a firm’s comparative advantage in basic research explaining its expansion into multiple industries. In an ideal setting, we would vary a firm’s economic scope in order to observe its impact on research patterns. To get as close as possible to this framework, we follow an instrumental variable approach and exploit an historical event that affected the economic scope of firms.

In 1981, Francois Mitterrand became president of the French Republic and implemented a vast nationalization program across industries. Even before his election, the tradition of French state intervention had resulted in a significant fraction of the economy being under state control. Consistent with Colbertist policies, the state modified the economic scope of firms by merging often unrelated firms into large conglomerates of national champions. [Cohen and Bauer \(1985\)](#) eloquently describe the phenomenon as a “game of monopoly” played by politicians. In 1987, the election of Jacques Chirac on a liberal platform marked the beginning of privatizations, which then continued into the 1990s under the impetus of the European Commission’s competition directorate. Using historical data on ownership structures, we define state participation in 1985-1987 as our instrumental variable.

The first-stage estimates reported in columns (1) to (3) of Table 3 confirm that state participation in the 1980s is associated with, on average, 1.5 more industry links for firms between 2000 and 2006. The associated F-test is well above the critical levels related to weak instruments. The second-stage estimates, reported in column (4), suggest that the instrumented estimate is significantly larger than the OLS one.²¹ This would suggest that our estimates might be a relatively conservative quantification of spillovers.

The exclusion restriction associated with our identification strategy requires that state participation in the 1980s should not directly affect a firm’s research incentives in the year 2000. To strengthen the argument, all of our IV specifications include a measure of continued state participation in the ownership of the firm. Another possible objection is that past ownership by the state also affects the likelihood of accessing subsidies and government contracts. In columns (2) and (5), we include

²⁰We also re-estimated the spillover parameter excluding an outlier industry in terms of basic research intensity that we identify in the Appendix. Estimates were unaffected.

²¹The fact that the OLS estimate represents a lower bound to our spillover parameter is consistent with agency theories in financial economics that highlight the difficulty of capital allocation in diversified firms and the associated valuation discount ([Jensen and Meckling, 1976](#); [Rajan, Servaes, and Zingales, 2000](#); [Scharfstein and Stein, 2000](#)).

TABLE 3: BASIC RESEARCH AND MULTI-MARKET ACTIVITY - INSTRUMENTAL VARIABLES

	(1)	1st Stage (2)	(3)	(4)	2nd Stage (5)	(6)
State Participation in 1985	.293*** (.056)	.288*** (.057)	.264*** (.056)			
Log # of Industries				0.085*** (0.018)	0.079*** (0.021)	0.070** (0.028)
Log Employment	.120*** (.003)	.120*** (.004)	.117*** (.004)	-0.007** (0.003)	-0.006 (0.004)	-0.006 (0.004)
State Participation	-.057 (.083)	-.057 (.083)	-.049 (.082)	-0.003 (0.011)	-0.005 (0.012)	-0.002 (0.013)
Public R&D Subsidies		.013* (.008)			0.010*** (0.004)	
Outsourcing to Univ.			.067*** (.012)			0.019** (0.010)
Year & Organization FE	Yes	Yes	Yes	Yes	Yes	Yes
Areas of Research Shares	Yes	Yes	Yes	Yes	Yes	Yes
N	13707	13707	13707	13707	13707	13707

Notes: Pooled data for the period 2000-2006. Estimates are obtained using Tobit models and relate to the marginal effect of the variables at the sample mean. *Basic Research Intensity* is defined as the ratio of total firm investment in basic research divided by total firm investment in applied research. *Log # of Industries* is the number of distinct SIC codes in which a firm is present. *State Participation in 1985* is a binary variable equal to 1 if the French state was a shareholder of the firm in 1985. *State Participation* is a binary variable equal to 1 if the French state is currently a shareholder of the firm. *Public R&D Subsidies* is a binary variable equal to 1 if the firm obtained public R&D subsidies. *University Collaboration* is a binary variable equal to 1 if the firm collaborates/outsources some of its R&D to the public system. *Log Employment* is the total employment of firms. *Share in Software/Biotech/Materials* is the share of R&D investment in the respective areas of research. *Industry FE* denote fixed effects for 1-digit SIC activities. *Year FE* denotes year fixed effects, and *Organization FE* denotes whether the firm operates its activity as a conglomerate or as a business group. See the Appendix for the definition of variables. Robust standard errors clustered at the firm level are in parentheses. One star denotes significance at the 10% level; two stars denote significance at the 5% level; and three stars denote significance at the 1% level.

as a control variable the share of public funding obtained by these firms. While public funding of private research is associated with higher basic research, its effect does not seem to confound our IV estimate related to the spillover. In columns (3) and (6), we again exploit the R&D survey to construct a dummy variable for links between private companies and universities. Again the estimates remain very similar and seem to exclude a differential access to the knowledge created by the public university system.

Taken together, the estimates across these specifications confirms the robustness of the correlation between firms' multi-industry presence and their investment in basic research.²²

²²Table 2 in Appendix B further addresses concerns about the measurement of the cross-industry spillover. Column (1) presents the benchmark specification. In column (2), we recompute multi-industry presence and size by excluding links to banks, insurance companies, and asset management companies. In addition, we explicitly control for the presence of financial activities within the group (either as a subsidiary or as the head of the group). Column (3) recomputes multi-industry presence and size by considering only activities with at least ten employees. We also add two variables in the specification that should capture the possibility of spurious industry presence. The first is a dummy variable on whether the headquarters are foreign. The second is a Herfindahl index related to the concentration of employment within a given industry of the firm's portfolio. Column (4) uses the data on the population of French firms and measures the frequency of

3.3 Citation Patterns Across Research Types

We next use citation information from the NBER patent data to empirically measure the quality step size for basic and applied innovation, as well as their importance for follow-up innovations (Hall, Jaffe, and Trajtenberg, 2001). An empirical issue is how to distinguish between patents derived from basic and from applied research. Trajtenberg, Henderson, and Jaffe (1997) document that patents from the public sector are more basic-research oriented than patents from the corporate sector. Consequently, we use the assignee code to distinguish between patents applied for by corporations from patents applied for by public institutions.²³ The evidence on citation patterns across public- and private-sector patents is organized in Table 4 and explained in Figure 6.

In Panel A of Table 4, we provide a first breakdown of public and private patents according to the number of direct, 1st generation citations that the originating patent received. In the model, the citation distributions for both public and private patents are directly sensitive to the underlying step sizes. The larger the step size, the more citations a patent receives, on average. We find that, on average, a public patent obtains 1.12 more citations than corporate patents.

To estimate the cooldown parameter, we proceed in several steps. First, we again consider whether the originating patent is public or private. Second, for the estimation of the cooldown parameter, instead of computing the 1st generation direct citations (middle panel of Figure 6) received by the original patents, we measure the importance of these 1st generation patents through the number of patent citations they received in turn from 2nd generation patents (bottom panel of Figure 6).²⁴ We do so for each year since the granting date of the original patent. Panel B reports the mean difference of this measure between patents originated in the public sector and patents originated in the private sector. We find that the data display a pattern whereby owners of patents following/citing a public-sector-originated patent within a few years are of higher quality than those citing private-sector-originated patents. The difference in the quality of follow-up patents oscillates between 0.3 to 0.6 citations in the early years. However, it drops significantly in the 8th year. The results are similar when using the Wilcoxon-Mann-Whitney test. The second row adjusts for the fact that patents from basic research have a broader applicability. Therefore, we use the concordance table developed by Silverman (1996), linking the International Patent Classification (IPC) system to the U.S. Standard Industrial Classification (SIC) system. This concordance enables us to condition citations on being within the same SIC code and qualitatively confirms the previous pattern.²⁵

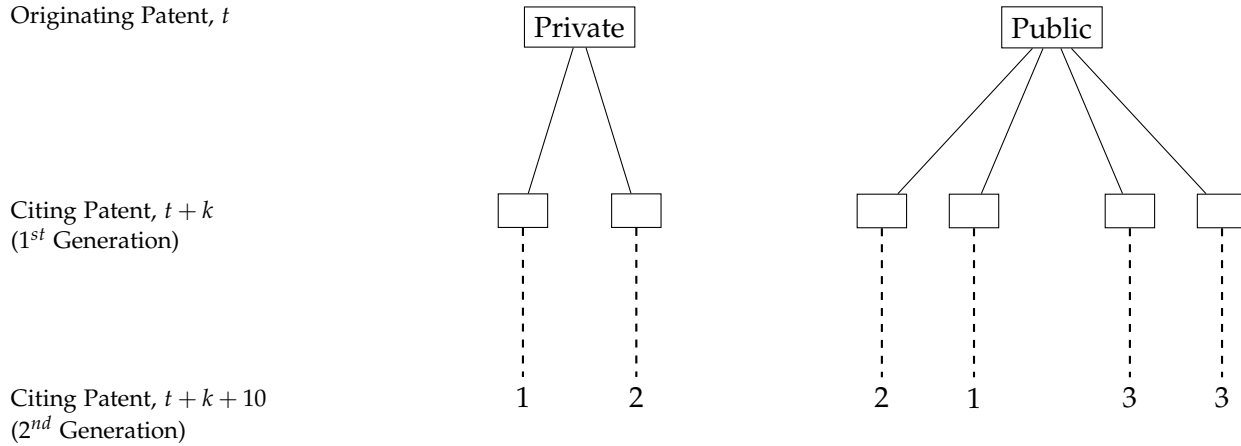
each distinct activity pair. We then inversely weight the distinct bilateral links of our innovative firms. All estimates remain robust.

²³The use of US patent data was linked to the availability of a long time horizon of publicly available data on patents granted, depositor classification, and the associated citations. The analysis of our final dataset will focus exclusively on French patenters, but the construction of the different variables will use information from the entire dataset. While our proxy is simple to measure in the data, it potentially misclassifies the contribution of private basic patents. However, given that our interest lies in the relative difference between those two groups of patents across time, time-invariant errors in the classification should not impact our conclusions. Appendix C provides additional robustness checks for the estimates on the cooldown rate of patents originating from basic and applied research.

²⁴All citation measures are based on cumulative 10-year citation horizons to avoid truncation effects.

²⁵It is likely that the use of concordance tables introduces some degree of measurement error, and explains why the citation patterns is more noisy relative to the previous row.

FIGURE 6: CONSTRUCTION OF CITATION INFORMATION



One potential limitation of the above analysis is that the propensity to patent research could be different between the public and private sector and might lead us to mismeasure the role of basic research. Unfortunately, such a differential pattern is hard to quantify with the current patent information and would require new micro-data to be collected. Hopefully our paper will spur the collection of such data and allow a better understanding of the production function of the public sector.

TABLE 4: PATENT CITATIONS ACROSS PUBLIC AND PRIVATE PATENTS

		Panel A: 1 st Generation Patents							
		Mean	25th	Median	75th	St. Dev.	Min	Max	N
Private Citations		5.88	1.00	4.00	7.00	8.01	0.00	179.00	28247.00
Public Citations		7.01	2.00	4.00	8.00	11.86	0.00	351.00	2806.00

		Panel B: 2 nd Generation Patents									
Age Cohort		1	2	3	4	5	6	7	8	9	10
Mean Difference		.3**	.3**	.62***	.28**	.41**	.23	.71***	.08	.39	.14
		(0.15)	(0.15)	(0.17)	(0.14)	(0.18)	(0.17)	(0.25)	(0.16)	(0.25)	(0.24)
Adjusted		.3**	.23*	.55***	.26*	.46**	.18	.58**	.06	.47*	.22
		(0.15)	(0.14)	(0.17)	(0.17)	(0.17)	(0.17)	(0.24)	(0.15)	(0.26)	(0.28)

Notes: Citation patterns of patents granted by the USPTO to French private and public depositors. Panel A computes direct 1st generation citations across public and private patents. Panel B reports the average difference in citations of 2nd generation patents for the period 1975-1990. Adjusted citations are conditioned on being within the same 1-digit industry of use, as indicated by the concordance table in Silverman (2004). Two sample t-tests with unequal variances were used. One star denotes significance at the 10% level; two stars denote significance at the 5% level; and three stars denote significance at the 1% level.

4 Quantitative Analysis

4.1 Estimation

In this section, we describe the estimation strategy used. We assume that the idiosyncratic fixed-cost component of basic research is drawn from a lognormal distribution with $\log(\phi) \sim \mathcal{N}(\bar{\phi}, \sigma^2)$. As a result, the set of parameters of the model is

$$\theta = \{\delta, \gamma, \varepsilon, p, \eta, \lambda, E, U, v_a, v_b, \zeta_a, \zeta_b, \kappa, \bar{\phi}, \sigma, \zeta, \varsigma\} \in \Theta.$$

In our dataset, for each firm f and each time period t , we have a vector of N observables from the actual data $\mathbf{y}_{ft} \equiv [y_{ft}^1 \cdot \cdot \cdot y_{ft}^N]_{N \times 1}'$ that includes the number of industries in which the firm is present, sales, profits, and labor costs. Let the entire dataset be denoted by $\mathbf{y}$.

We use the simulated method of moments (SMM) for the estimation.²⁶ Define $\Lambda(\mathbf{y})$ and $\Lambda(\theta)$ to be, respectively, the vectors of real data moments (generated from $\mathbf{y}$) and equilibrium model moments (generated for some vector of parameters θ). Since certain moments require knowledge of the joint distribution of firms over the number of products and industries (m, n) and the portfolio of product qualities $\mathbf{q}$, which has no apparent analytic form, we simulate a large panel of firms to calculate $\Lambda(\theta)$ to a high degree of accuracy.²⁷

Our proposed estimator minimizes a quadratic form of the difference between these two vectors

$$\hat{\theta} = \arg \min_{\theta \in \Theta} [\Lambda(\theta) - \Lambda(\mathbf{y})] \cdot W \cdot [\Lambda(\theta) - \Lambda(\mathbf{y})], \quad (23)$$

where W is the weighting matrix. We use a diagonal weighting matrix with entries equal to the inverse square of the data moment value, or, in notational terms, $W_{ii} = 1/\Lambda_i(\mathbf{y})^2$ and $W_{ij} = 0$ for $i \neq j$. In our estimation, we use 30 moments, which we denote as $\Lambda(1) - \Lambda(30)$. We pick moments that are most informative for identification of the unique features of our model.

Status Quo Policies We also take into account that, during the period we consider, there was existing government support for R&D activities in France. In our dataset, 10% of corporate R&D is, on average, publicly funded. Therefore, in our estimation, we introduce a uniform subsidy to the total R&D spending of the firm $\psi_a = \psi_b = 0.10$. The government has a balanced budget every period, so that the sum of total subsidies (S) and public research funding (R) must be equal to tax revenues; that is,

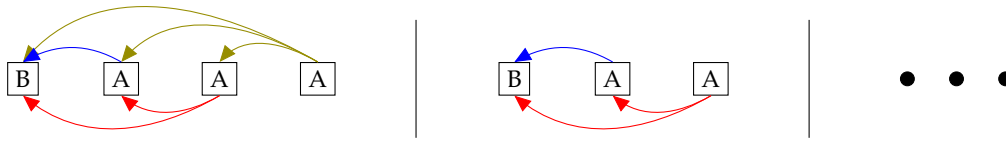
$$T = S + R = \psi_a h_a(a) + \psi_b \sum_{m=1}^M \mu_m P_m [h_b(b_m) + \bar{\phi}_m] + U [h_b(u) + \bar{\phi}],$$

where T is a lump-sum tax on consumers. In France, during 2000-2006, the fraction of GDP devoted to public research labs and academic institutions was approximately 0.5%. Therefore, we pick R/Z , which is the share of GDP devoted to public basic research, to be 0.5%.

²⁶See Bloom (2009) and Lentz and Mortensen (2008) for further description and usage information on SMM. Appendix E provides additional details about the computer algorithm.

²⁷For our results, we simulate 32K firms with a burn-in time of 100 years and 100 time steps per year.

FIGURE 7: CITATION MODEL DIAGRAM



Basic innovations are represented by “B”, while applied innovations are represented by “A”. Each arrow represents the possibility of a citation. An actual citation will occur with probability ηx for basic innovations and λx for applied innovations.

4.2 Identification

Ultimately, the parameters of the model are identified jointly by their optimality with respect to the estimation objective function defined above. Nonetheless, there is a clear sense in which certain parameters are identified mainly through certain moments. In this section, we provide an account of such relations with special focus on those parameters that are either novel to our model or are particularly important for the ensuing policy exercises. Additionally, we provide the full Jacobian matrix of the moment vector Λ with respect to the parameter vector θ . This provides precise numerical values for the direction and magnitude of moment-parameter dependence in the neighborhood of our estimated parameter values.

4.2.1 Step Sizes: Citation Model

The innovation step sizes for basic and applied research are two of the most critical parameters in the model, as they fix the scale of innovation sizes against which input costs are weighed. Though these values are important in determining the overall growth rate, profitability of firms, and the firm-size distribution, the distribution of patent citations provides the most direct evidence. In the model, the citation distributions for both public and private patents are directly sensitive to the underlying step sizes. The larger the step size, the more citations a patent receives, on average.

The distinction between basic and applied research has a very natural interpretation in this setting. We interpret each basic innovation as starting a new line of research that is cited by subsequent applied innovations that build off of it. When a subsequent basic innovation is generated, the original basic patent receives no more citations, and the new line of research prevails. Similarly, applied innovations also build off of one another, and we allow them to cite prior patents both from basic and applied research. We model the probability of being cited by subsequent innovations as some parameter x times the step size of the innovation. This encapsulates the notion that patents with a high immediate impact on productivity are also generally more highly cited.

Recall that the two step sizes in our model are η and λ , and that we have applied and basic innovation rates τ_b and τ_a . We show in Appendix A that under the dynamics described above, the total number of citations a patent receives will follow a geometric distribution. In particular, the

probability of a patent with step size s receiving n citations will be

$$p_s(n) = (1 - c_s)c_s^n \quad \text{where} \quad c_s = \frac{\tau_a x s}{\tau_a x s + \tau_b}.$$

Additionally, we know that basic innovations always have step size η , while applied innovations have step size η when they land on “hot” product lines (with probability α) and step size λ otherwise. Thus, we find that

$$\begin{aligned} p_b(n) &= p_\eta(n) \\ p_a(n) &= \alpha p_\eta(n) + (1 - \alpha)p_\lambda(n). \end{aligned}$$

This structure has the advantage that it not only generates average citation rates and counts for basic and applied research, but also generates an entire predicted distribution of citation counts across patents. Importantly, these distributions are directly sensitive to the underlying step sizes. The larger the step size, the more citations a patent receives, on average. Since the scale parameter x is essentially setting units, this soaks up one degree of freedom. But the remaining difference between basic and applied citations is informative about the difference between the respective step sizes. Meanwhile, τ_b and τ_a are constrained through equilibrium by the firm’s dynamics moments, while the absolute step size is constrained by the aggregate growth rate.

While we don’t observe patents as being basic or applied directly in the data, we can proxy these differences through the public/private origin of the assignee. Our model yields predictions for mean and root-mean-squared (RMS) citations for both public and private patents separately. We use these four moments in the main estimation. Even though they are not targeted directly, we also find that the distributions of patent citations for both private and public patents match those seen in the data quite well. As seen in Figures 8 and 9, this close match gives us confidence that the way in which we model citations is, indeed, appropriate for this setting.

FIGURE 8: PRIVATE CITATION DISTRIBUTIONS

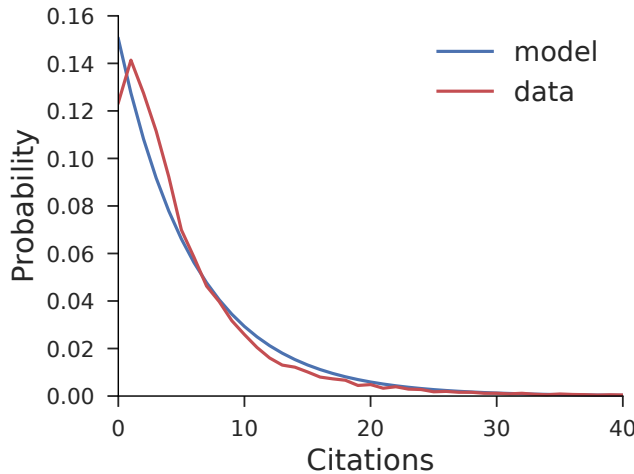


FIGURE 9: PUBLIC CITATION DISTRIBUTIONS

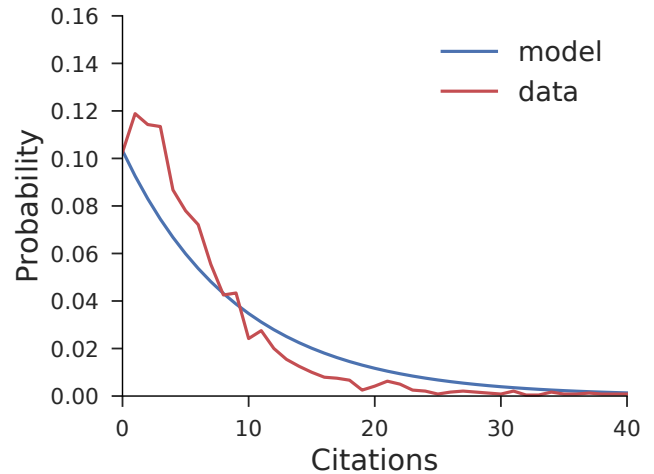
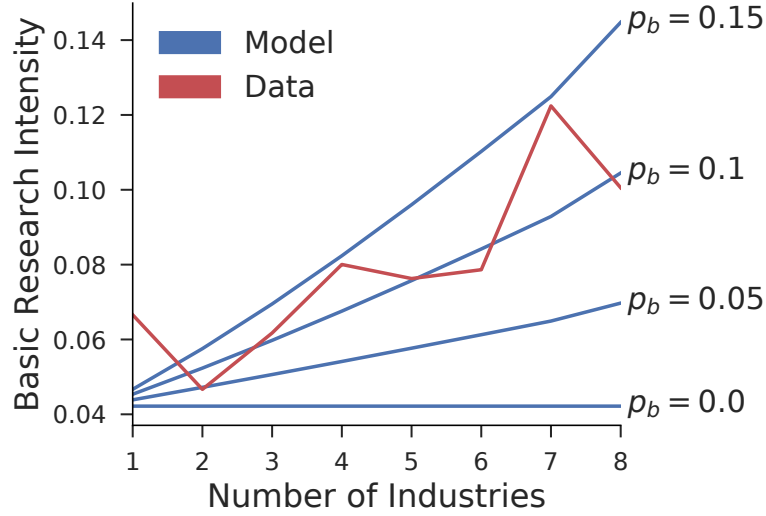


FIGURE 10: BASIC RESEARCH INTENSITY VS SPILLOVER PARAMETER p 

4.2.2 Across-industry Spillovers

We define basic research intensity as the ratio of spending on basic research to spending on applied research. Since the effect of multi-industry presence on this quantity is of critical importance to our model, we have one moment for each $\hat{m} \in \{1, \dots, M\}$, the number of industries a firm is observed to be operating in. Note that firms in the model might be “present” in an industry but currently have no products there, meaning the observed $\hat{m}$ might be different from the underlying m . Given a set of parameters and an equilibrium of the model, this moment’s value for a given $\hat{m}$ is

$$\Lambda(1 - 8) = \mathbb{E}_m \left[\frac{P_m(h_b(b_m) + \bar{\phi}_m)}{h_a(a_m)} \mid \hat{m} \right].$$

In our estimation, we use $M = 10$. However, in the data, there are only a handful of firms with $\hat{m} > 8$, so we have one moment for each $\hat{m} \in \{1, \dots, 7\}$ and a final moment that is averaged over $\hat{m} \in \{8, 9, 10\}$. The way in which this moment increases with $\hat{m}$ identifies the cross-industry spillover parameter p in our model. Additionally, the overall level provides us with some identification power for the basic research cost parameters (ξ_b, ν_b) .

Figure 10 plots the empirical (red line) and model counterparts for these moments for different spillover parameter $p = \{0.00, 0.05, 0.10, 0.15\}$. The figure makes it clear that our cross-industry spillover parameter p is identified by matching the basic research intensity of multi-industry firms.

The slope of the analogous graph for the extensive margin of basic research (P_m) is also sensitive to the magnitude of cross-industry spillovers. Thus, we use the share of positive basic research spending by each $\hat{m}$ to provide additional information on p . The overall level of positive basic research is also directly informative about the mean $\bar{\phi}$ and variance σ^2 of the fixed-cost component of basic research. Analytically, the probability that a firm with a certain industry presence $\hat{m}$ undertakes basic research is simply the probability that the idiosyncratic fixed-cost draw is less than the cutoff for that value of $\hat{m}$

$$\Lambda(9 - 16) = \mathbb{E}_m [P_m \mid \hat{m}].$$

Though not plotted here, one can see in the Jacobian matrix in Table 4 of Appendix F.2 that the average fixed-cost parameter $\bar{\phi}$ strongly influences the level of this moment for all values of $\hat{m}$. Furthermore, the fixed-cost variance parameter σ induces a tilt in these moments across different $\hat{m}$ values. That is to say, increasing σ raises the probability of performing basic research for low $\hat{m}$ firms and lowers it for high $\hat{m}$ firms. This is consistent with the intuition of the model and with the properties of the lognormal distribution.

4.2.3 Within-industry Spillovers

As discussed in Section 3.3, we use patent citation data to quantify the within-industry spillovers associated with basic research. The model predicts that innovations that build off of previous basic research should have, on average, a larger step size. In this setting, innovations with a larger step size also see more citations. Thus, patents that cite basic innovations should themselves have more citations. We empirically confirmed that the magnitude of this effect diminishes with the age of the patent due to product line cooldown ζ .

Consider a public innovation in the model: this will generate a public patent and make a particular private product line hot for some time. During this time, any subsequent (citing) innovations, whether basic or applied, will yield a basic innovation step size η , just as all public patents do. This step size will determine how many citations these citing patents garner down the line. Then, at some point in time, the product line will cool down and again begin yielding step sizes determined by the research type, either λ or η . Thus, the average cooldown time is also the average time after which a public innovation is indistinguishable from a private innovation, which is given by

$$\Lambda(24) = \frac{1}{\zeta} \left(\frac{\tau_a}{\tau} \right).$$

The first term arises due to the standard properties of a Poisson process with rate ζ , while the second term arises from the fact that another basic innovation may hit the product line before cooldown occurs, in which case the original line would receive no further citations. This yields direct information on the value of the cooldown rate ζ , as seen in the Jacobian matrix.

4.2.4 Industry Expansion

We track two moments relating to the distribution of $\hat{m}$, the mean and mean squared. They are given by

$$\Lambda(17) = \mathbf{E}_{\hat{m}} [\hat{m}] \quad \text{and} \quad \Lambda(18) = \mathbf{E}_{\hat{m}} [\hat{m}^2].$$

Looking at the Jacobian matrix, one can see that the mean m value and the mass of potential entrants are closely linked, as successful entrants start with the lowest working knowledge value $m = 1$. Similarly, the mean of m^2 is strongly influenced by the entrant buyout probability parameter ς . Intuitively, when entrants are more likely to be bought out by incumbents, which results in industry expansion with probability z , this will concentrate industry expertise in existing firms, thus increasing the dispersion of m .

Empirically, we match the distribution of firms across industry classes, as shown previously in the left panel of Figure 4.²⁸

4.2.5 Remaining Moments and Parameters

The remaining moments used in the estimation of the model include firm-level profitability, employment growth, age, and total research intensity. At the aggregate level, we use the growth rate of the French economy during the sample period. In Appendix F, we provide a detailed discussion of the mapping between these moments and the parameters of the model.

4.3 Estimation Results

Table 5 reports the values of the estimated structural parameters. The estimated values of the discount rate and CRRA utility parameters are within their standard macro ranges. The elasticity of substitution parameter generates 17%(= $1/\varepsilon$) gross profits, resulting in 7.9% net profits after subtracting R&D expenses as a share of sales.

TABLE 5: PARAMETER ESTIMATES

#	Description	Sym	Value	#	Description	Sym	Value
1.	Discount Rate	δ	0.039	10.	Applied Cost Curvature	ν_a	1.367
2.	CRRA Utility Parameter	γ	3.020	11.	Basic Cost Curvature	ν_b	1.539
3.	Elasticity of Substitution	ε	5.863	12.	Applied Cost Scale	ζ_a	1.230
4.	Cross-industry Spillover	p	0.116	13.	Basic Cost Scale	ζ_b	5.445
5.	Buyout Rate	ς	0.456	14.	Exogenous Exit Rate	κ	0.006
6.	Basic Step Size	η	0.079	15.	Basic Fixed Mean	$\bar{\phi}$	-4.759
7.	Applied Step Size	λ	0.050	16.	Basic Fixed Std. Dev.	σ	0.329
8.	Mass of Entrants	E	0.495	17.	Product Cooldown Rate	ζ	0.117
9.	Mass of Academic Labs	U	0.493	18.	Citation Rate	x	2.984

One of the most important parameters of our model is the cross-industry spillover parameter $p = 0.12$, which measures the probability that a basic innovation will have any additional immediate applications (it may have many). This estimate affects the extent to which basic innovations contribute to cross-sectional growth. In equilibrium, firms operate in two out of ten industries on average. Therefore, any given spillover is not embodied with probability 89%(= $8/9$). Given that the probability of having a spillover is 12%, the probability of having a disembodied spillover is 10%(= $0.12 * 0.89$).

The estimated innovation size of basic research is $\eta = 7.9\%$, and the innovation size of each new applied innovation is $\lambda = 5.0\%$. This implies that basic research (hot product lines) makes applied innovation 58%(= $7.9/5.0 - 1$) more productive.

²⁸As discussed in Section 3.2, there are both costs and benefits to multi-industry presence. In our setting, a broad economic scope allows firms to better internalize spillovers from research. At the same time, the corporate finance literature has highlighted the difficulty of capital allocation in diversified firms. Thus, the empirical distribution that we target reflects the net trade-off between these forces.

Additionally, each basic innovation has a within-industry spillover. The cooldown rate of hot product lines is estimated to be $\zeta = 0.12$, which indicates that a basic innovation affects the subsequent innovations in the same product line for almost $8.5 (= 1/0.12)$ years, on average.

The elasticity of applied innovation counts with respect to the research dollars spent is estimated to be $0.73 (= 1/\nu_a)$, and, similarly, the elasticity of basic innovation with respect to the basic research investment is $0.65 (= 1/\nu_b)$. These values are close to the elasticity estimates in the literature, which typically find a value of around 0.5 (Griliches, 1990; Pakes and Griliches, 1984; Kortum, 1992; Kortum, 1993).

4.4 Goodness of Fit

Table 6 contains the moments from the actual data and our estimated model.

TABLE 6: MOMENTS USED IN ESTIMATION

#	Description	Model	Data	#	Description	Model	Data
$\Lambda(1-8)$	Basic Research Extensive	See Figure 12		$\Lambda(23)$	R&D/Labor	0.276	0.260
$\Lambda(9-16)$	Basic Research Intensive	See Figure 11		$\Lambda(24)$	Employment Growth	0.103	0.103
$\Lambda(17)$	Mean Industries	2.264	2.203	$\Lambda(25)$	Aggregate Growth	0.013	0.015
$\Lambda(18)$	Mean Square Industries	7.535	6.976	$\Lambda(26)$	Spillover Differential	8.344	8.000
$\Lambda(19)$	Return on Sales	0.033	0.033	$\Lambda(27)$	Public Citations Mean	8.682	7.013
$\Lambda(20)$	Exit Rate	0.092	0.092	$\Lambda(28)$	Public Citations RMS	12.63	14.20
$\Lambda(21)$	Age, Small Firms	11.77	15.00	$\Lambda(29)$	Private Citations Mean	5.747	5.885
$\Lambda(22)$	Age, Large Firms	19.22	24.87	$\Lambda(30)$	Private Citations RMS	8.581	9.154

The results indicate that the model performs very well in generating firm and industry dynamics similar to those in the data. Consistent with our data, a significant fraction of innovating firms invest in basic research: in our model, in particular, 29% of firms invest in basic research, while, in the data, 27% invest. We also capture the positive relationship between the extensive margin of basic research and multi-industry presence, as evidenced in Table 6 and Figures 11 and 12.

The positive correlation between a firm's multi-industry presence and its basic research intensity was one of the major predictions of our model. As explained previously, multi-industry presence plays an important role in increasing basic research incentives, by allowing a greater potential to internalize the positive spillovers from basic research. In our reduced-form analysis, we confirmed the significant and positive correlation between multi-industry presence and basic research intensity. This has been the key moment to identify the cross-industry spillover parameter. Our model successfully generates this positive correlation.

In the data, firms operate, on average, in 2.2 industries, and the same is roughly true in the model. Furthermore, we find the mean squared in the model to be 7.5, compared to 7.0 observed in the data.

In addition, the mean profitability is 3.2% in the model and in the data. The prime determinants of profitability are the step sizes for basic and applied innovation. However, these also affect the investment levels for both types of research, since this increases the return to successful innovation. Therefore, the step-size parameters are set to be a compromise between hitting the profitability

FIGURE 11: FRACTION POSITIVE BASIC BY # INDUSTRIES

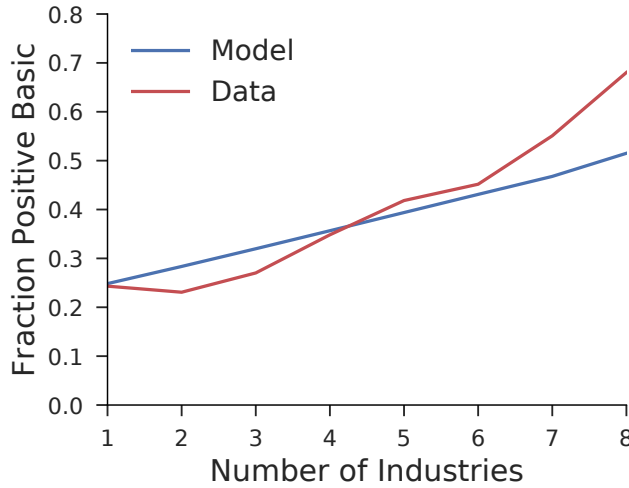
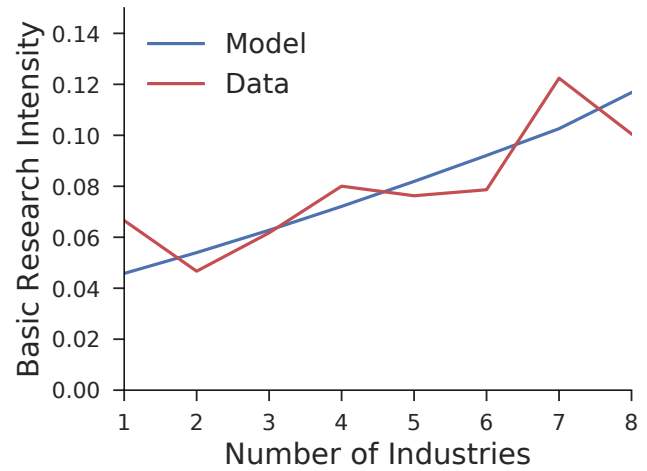


FIGURE 12: BASIC RESEARCH INTENSITY BY # INDUSTRIES



moment and the research investment and growth moments.

All of these components of the economy determine the aggregate growth rate. Our model matches the observed growth rate closely. Our model economy grows at a rate of 1.3%, while the French economy grew at an average rate of 1.5% during the period studied (2000-2006).²⁹

5 Policy Analysis

In this section we first study efficiency properties of the decentralized equilibrium in Section 5.1, and the social planner solution in Section 5.2. We then turn to various policies that could address these inefficiencies. Section 5.3 considers a uniform research subsidy that currently exists in many OECD economies. Given our distinction between basic and applied research, we propose in Section 5.4 a hypothetical policy that subsidizes the two types of research differently. This could potentially generate a measurement problem since firms would have an incentive to misreport the type of research they undertake. In Section 5.5, we explore how the scope for misclassification affects optimal policies and welfare. Section 5.6 focuses on the financing and role of the public sector in economic growth. Finally, in Section 5.7, we discuss the robustness of our policy conclusions to extensions of the model.

5.1 Status Quo: Baseline Economy

Table 7 provides equilibrium values for some of the important variables in the model:

In this table, τ_a denotes the aggregate rate of applied innovation by incumbents and entrants, whereas τ_b^e denotes basic innovation (embodied—i.e., internalized) by private firms, and τ_u denotes basic innovation done by academic research labs. The next five columns report the labor allocations

²⁹In Appendix F.3 we also discuss the model's prediction about moments that were not directly targeted. These include the correlation between firm profitability and basic research intensity, the correlation between multi-industry presence and firm size, and the skewness of the firm size distribution.

TABLE 7: DECENTRALIZED ECONOMY: ENDOGENOUS VARIABLES (IN PERCENTAGES)

ψ	R/Z	τ_a	τ_b^e	τ_u	L_p	L_b	L_u	L_e	L_a	α	g	β
subsidy rate	public basic in GDP	applied innovation	private basic innovation	public basic innovation	production labor	labor in private basic	labor in public basic	entrant labor	labor in private applied	share of hot product lines	growth rate	welfare
10	0.5	14.7	0.60	0.13	85.6	0.55	0.52	4.4	8.9	7.2	1.35	100.0

into production, private basic, public basic, entry, and applied research. The remaining columns report the fraction of hot product lines α , the growth rate g , and the welfare in consumption-equivalent terms β .

In our benchmark economy, 85.6% of labor is used for production, and 14.4% for innovation activities. Among researchers, roughly 7% are engaged in basic research activities. Note that this composition within innovation activities will be the central focus of the policy analysis, as uninternalized (potential) spillovers are one of the main sources of inefficiency. In order to study the welfare properties of this economy, we normalize the benchmark welfare to $\beta = 1.00$ and compare it to the social planner's optimum, which we will analyze next.

5.2 Quantifying the Social Planner's Optimum

Since our focus is on innovation policies, we consider a social planner who controls the R&D production in the economy while being subject to the same monopoly distortions on the production side (hence, we do not consider a production subsidy in our policy analysis either). Table 8 summarizes these results.

TABLE 8: SOCIAL PLANNER'S OPTIMUM (IN PERCENTAGES)

Policy	ψ_a	ψ_b	R/Z	τ_a	τ_b^e	τ_u	L_p	L_b	L_u	L_e	L_a	α	g	β
Baseline	10.0	10.0	0.5	14.7	0.6	0.13	85.6	0.6	0.52	4.4	8.9	7.2	1.35	100.00
Soc Plan	-	-	-	12.0	3.8	1.89	83.1	4.5	2.22	3.4	6.8	35.4	1.88	104.61

See Table 7 for variable definitions.

One striking feature of the solution to the social planner's problem under both scenarios is that the fraction of labor devoted to research activities is not substantially greater than in the decentralized equilibrium. In particular, the total labor allocated to production activities is 85.6% in the decentralized economy, while it is 83.1% when set by the social planner. Hence, there is a slight overinvestment in production labor relative to research labor, but this misallocation is quantitatively quite small.

Indeed, the dominant misallocation here is not between production and research, as is common in this class of models, but among the various types of research activities – in this case, applied and basic innovation. In the decentralized economy, only 1.07% of the total labor force is devoted to basic research, whereas in the social planner's economy, this number rises to 6.7%. In other words, the social planner devotes 40% of research labor to basic research, whereas this fraction is only 7% in the decentralized economy. This happens on both the intensive and the extensive margins of basic

research. In fact, the planner finds it optimal to employ nearly all private research labs, regardless of their fixed-cost draw.

Another interesting and important finding is that, in the case of applied innovation, there is actually an *overinvestment* in the baseline economy. The applied research labor utilization (including entrants) is 13.3% in the decentralized case. This figure drops to 10.2% in the social planner's solution. This is in spite of the fact that the fraction of hot product lines rises from 7% to 35%, meaning that the average step size of an applied innovation rises by almost a third.

The net result of the above changes is that growth rises from 1.35% to 1.88%. Overall, the decentralized economy's welfare corresponds to a decrease of 4.6% in consumption-equivalent terms from the social planner's optimum. The following policy experiments will try to bridge this gap.

5.3 Uniform Private Research Subsidy

Consider the case in which the government subsidizes a fraction ψ of each firm's total research investment. Note that such a policy subsidizes both basic and applied research, similar to the current R&D subsidy policy in the U.S. and around the world. Funding of academic research is kept unchanged. Table 9 summarizes the results of the optimal uniform subsidy rate.

TABLE 9: UNIFORM RESEARCH SUBSIDY (IN PERCENTAGES)

Policy	ψ_a	ψ_b	R/Z	τ_a	τ_b^e	τ_u	L_p	L_b	L_u	L_e	L_a	α	g	β
Baseline	10.0	10.0	0.5	14.7	0.6	0.13	85.6	0.6	0.52	4.4	8.9	7.2	1.35	100.00
Soc Plan	-	-	-	12.0	3.8	1.89	83.1	4.5	2.22	3.4	6.8	35.4	1.88	104.61
Uniform	29.7	29.7	0.5	16.7	1.5	0.10	82.1	1.5	0.49	5.3	10.7	13.4	1.68	100.64

See Table 7 for variable definitions.

Our analysis of the baseline economy and the planner's economy documented a slight underinvestment in research overall and a large misallocation between the different types of research. A uniform subsidy is, therefore, ill-suited to address these issues, as it cannot directly affect the allocation between research types, and any attempt to subsidize basic research will only worsen the overinvestment in applied research.

Under this policy, we allocate a larger fraction of the labor force to research relative to the social planner's economy. Overall, the researcher's share goes up to 18% from 14%. This new level is quite close to the corresponding level of 17% seen in the social planner's optimum, but the allocation of research labor between basic and applied research is still suboptimal. As a result, the economy grows at a lower rate (1.7%) than the social planner's economy grows (1.9%). The welfare gain from this policy is 0.6 percentage points, which is only 13% ($= 0.6/4.6$) as large as the gains associated with moving to the social planner's optimum.

Although the underinvestment in basic research is sizable, the uniform policy partially makes up for this, at the cost of worsening the overinvestment in applied research. The main lesson to be drawn from this is that when considering a uniform research subsidy, one should take into account

the negative welfare effects associated with oversubsidization of applied research. Finding a feasible method to differentiate between basic and applied research is essential to better innovation policies.

5.4 Type-Dependent Research Subsidy

Now consider the case in which the policymaker sets different subsidy rates for the different types of private research efforts. Let ψ_a and ψ_b denote the applied research and basic research subsidy rates, respectively. The share of GDP allocated to public research (R/Z) is kept constant by the policymaker. Table 10 reports the optimal subsidy rates and resulting equilibrium variables.

TABLE 10: TYPE-DEPENDENT RESEARCH SUBSIDY (IN PERCENTAGES)

Policy	ψ_a	ψ_b	R/Z	τ_a	τ_b^e	τ_u	L_p	L_b	L_u	L_e	L_a	α	g	β
Baseline	10.0	10.0	0.5	14.7	0.6	0.13	85.6	0.6	0.52	4.4	8.9	7.2	1.35	100.00
Soc Plan	-	-	-	12.0	3.8	1.89	83.1	4.5	2.22	3.4	6.8	35.4	1.88	104.61
Uniform	29.7	29.7	0.5	16.7	1.5	0.10	82.1	1.5	0.49	5.3	10.7	13.4	1.68	100.64
Targeted	11.1	48.6	0.5	12.7	4.4	0.11	83.5	5.1	0.50	3.5	7.3	29.9	1.71	102.81

See Table 7 for variable definitions.

Since the underinvestment is mainly in basic research, the optimal type-dependent subsidy dictates a much larger subsidy rate for it – namely, $\psi_b = 49\%$ and $\psi_a = 11\%$. Here, the major component of policy is a more than fivefold increase in the subsidy rate for basic research, whereas the subsidy rate for applied innovation is virtually unchanged.

The large value for the basic research subsidy is straightforward to understand. There are spillovers associated with basic innovation that are not internalized by firms. By subsidizing this type of innovation, we can mitigate this effect. Contrary to the results seen in the uniform policy case, this policy can achieve a large fraction of the welfare gains seen in the social planner's optimum: 61% ($= 2.8/4.6$). Still, there is a good deal of room left for improvement, and in the next section, we investigate the sources of these remaining inefficiencies.

5.5 Misclassification of Research Type

The previous discussion highlights that the most promising policy tool to increase welfare is based on a type-dependent subsidy. One concern, however, is that it is not always clear whether or not a specific undertaking should be considered basic research. This might lead firms to misclassify research investment in order to exploit the differential subsidy rates. In this section, we quantify how optimal policies and welfare depend on the degree of misclassification of research.

In this setting, some portion of applied research spending might be misclassified as basic research and, therefore, obtain a more favorable subsidy rate. We can capture this theoretically by assuming that some fraction $z \in [0, 1]$ of applied research is misclassified as basic research. Letting v_a and v_b stand for the total present value associated with successful innovation of the respective type, this

results in the modified product-line level optimization

$$\Pi = \max_{x_a, x_b} x_a v_a + x_b v_b - (1 - [(1 - z)s_a + zs_b]) \tilde{w}c_a(x_a) - (1 - s_b) \tilde{w}c_b(x_b) \quad (24)$$

Thus, setting real subsidy rates s_a and s_b in this modified regime (characterized by the level of misclassification z) is equivalent to setting subsidy rates $\tilde{s}_a$ and $\tilde{s}_b$ in the regime without misclassification ($z = 0$), where these effective rates are given by

$$\tilde{s}_a = (1 - z)s_a + zs_b \quad (25)$$

$$\tilde{s}_b = s_b. \quad (26)$$

Given an optimal type-dependent subsidy scheme $\hat{s}_a$ and $\hat{s}_b$ from the case with no misclassification of research, we can exactly implement this by setting $\tilde{s}_a = \hat{s}_a$ and $\tilde{s}_b = \hat{s}_b$, which, in terms of real subsidy rates means

$$s_a = \frac{\hat{s}_a - z\hat{s}_b}{1 - z} \quad \text{and} \quad s_b = \hat{s}_b \quad (27)$$

As this expression makes clear, in the region where the desired s_a and s_b are both positive, we can replicate the optimal policy. However, if we are constrained to set positive subsidy rates, the full optimum cannot be implemented. This will be the case when the misclassification parameter exceeds a certain threshold—i.e.,

$$z > z^* \equiv \frac{\hat{s}_a}{\hat{s}_b}. \quad (28)$$

The optimal policies are illustrated in the left panel of Figure 13, where the red line indicates the true applied subsidy rate (s_a); the green line is the effective applied subsidy rate ($\tilde{s}_a$); and the blue line is the true basic subsidy rate that also corresponds to the effective one ($s_b = \tilde{s}_b$). The dashed vertical line represents the threshold at z^* . When the scope for misclassification is small (i.e., $z < z^*$), we can implement the optimal policy by setting the basic subsidy to the exact targeted rate ($s_b = \hat{s}_b$) and imposing some reduction on the applied research subsidy rate. When the misclassification level exceeds the threshold and a tax on applied research is not possible, we cannot restore the optimal policy ($\hat{s}_a, \hat{s}_b$) but can maximize welfare by setting the applied subsidy to 0 and finding a second-best policy level for s_b .

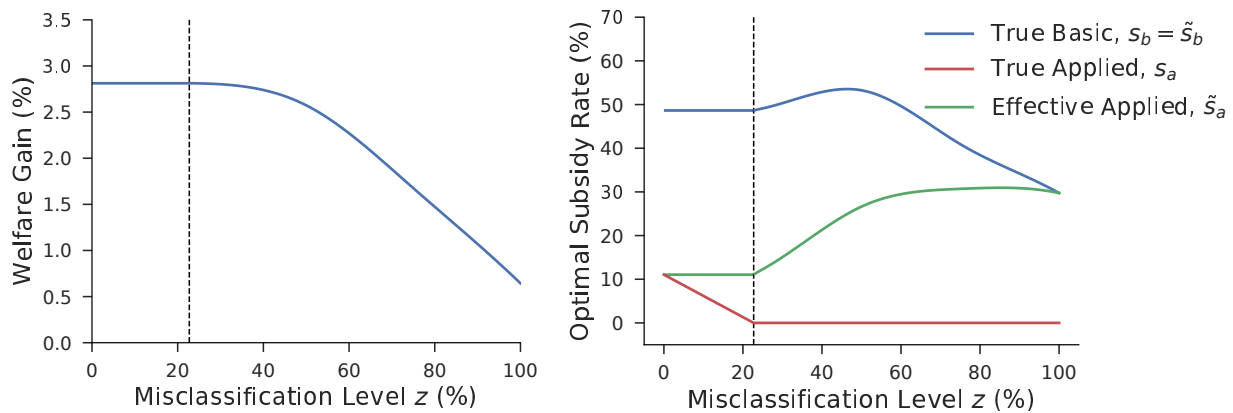
The welfare level achieved for each level of misclassification is plotted in the right panel of Figure 13. Importantly for policymakers, the welfare gains arising from a type-dependent subsidy are robust to a substantial amount of misclassification before falling. Loosely speaking, the welfare gains associated with a type-dependent research subsidy are roughly the same for any level of misclassification below 50%.³⁰

5.6 Academic Policy

An important question is whether and how the presence of a public research sector affects the optimal policy choices of the social planner. One possibility is that the public sector provides a more targeted

³⁰Chen, Liu, Suárez Serrato, and Xu (2018) analyze the effects of a Chinese policy that awards substantial corporate tax cuts to firms that increase R&D investment over a given threshold. They find, amongst others, that about 30% in the resulting increase of R&D is due to relabeling of administrative expenses.

FIGURE 13: OPTIMAL POLICIES & WELFARE WITH MISCLASSIFICATION OF RESEARCH



policy tool and is, therefore, a substitute for incentivizing private basic research efforts. Tables 11 and 12 explore these policy choices in several steps.

TABLE 11: OPTIMAL ACADEMIC POLICIES (IN PERCENTAGES)

Policy	ψ_a	ψ_b	R/Z	τ_a	τ_b^e	τ_u	L_p	L_b	L_u	L_e	L_a	α	g	β
Baseline	10.0	10.0	0.5	14.7	0.6	0.13	85.6	0.6	0.52	4.4	8.9	7.2	1.35	100.00
Soc Plan	-	-	-	12.0	3.8	1.89	83.1	4.5	2.22	3.4	6.8	35.4	1.88	104.61
Uniform	29.7	29.7	0.5	16.7	1.5	0.10	82.1	1.5	0.49	5.3	10.7	13.4	1.68	100.64
Targeted	11.1	48.6	0.5	12.7	4.4	0.11	83.5	5.1	0.50	3.5	7.3	29.9	1.71	102.81
Unif Acad	29.8	29.8	0.68	16.7	1.46	0.27	81.9	1.47	0.67	5.3	10.7	15.4	1.70	100.70
Unif Acad+	24.3	24.3	3.24	15.5	0.89	2.80	81.7	0.86	3.19	4.7	9.6	24.4	1.88	102.88

See Table 7 for variable definitions.

In the fifth row of Table 11, we allow the social planner to optimize not only on the uniform research subsidy to firms, but also on funding for the academic sector. The optimal funding for the academic sector calls for an increase in public investment by 0.18% of GDP, or a 40% increase with respect to the observed levels. Importantly, the optimal uniform subsidy to firms remains unchanged and suggests a potential complementary role for research policies targeted towards the public sector. Overall, the gains from moving to optimal academic policy are fairly modest and may reflect the delay in the applicability of academic research.

To explore this possibility, the sixth row of Table 11 relaxes the “Ivory Tower” assumption. In our model, academic research is assumed not to be immediately applicable to production. Its effects are borne out only through the effects of within-industry spillovers, with larger step sizes for applied innovations that build on it. Thus, there is a sense in which academic research is less immediately effective than private basic research, where innovations are directly used in production and spillovers are internalized when they fall within a firm’s industrial scope. So we also include the results of an additional counterfactual exercise in which academic innovation is immediately applicable (Unif Acad+), putting it more on par with private basic research. Here we can see that the optimal level of

academic funding goes up substantially, as do the associated welfare gains.³¹

Table 12 relaxes the assumption that the private and public research sectors have the same production functions. Unfortunately, we do not have data for our setting on the breakdown of costs of public and private research. However, the fact that there is a linkage between private and public research wages is a key insight that we exploit in this robustness check.³² Consequently, we simulate the model with different levels of public research productivity, with cost parity as a lower bound on the productivity of public researchers relative to private sector researchers. For each level of the productivity multiplier, we recompute the optimal uniform subsidy plus the academic spending-level policy, keeping other parameters the same as in the baseline estimation. Table 12 summarizes the results of this exercise.

TABLE 12: OPTIMAL SUBSIDIES WITH VARIOUS ACADEMIC PRODUCTIVITIES

Academic Multiplier	Optimal Uniform Subsidy	Optimal Academic Spending	Welfare Gain
1x	29.8	0.68	0.7
2x	29.2	1.01	1.0
5x	27.6	1.54	1.9

Increasing the productivity of public research increases the optimal level of funding given to public research in the form of academic spending and substantially increases the associated welfare gains. The optimal uniform subsidy remains nearly constant, decreasing only slightly. There are two primary and countervailing forces at play. The first is that more academic basic research will increase the number of “hot” product lines, thus increasing the returns to applied research by private firms. The second is that more academic spending on researchers will increase the wage in general equilibrium, thus increasing the costs of applied research. Our estimates suggest, therefore, that the optimal uniform subsidy remains a robust complementary tool with respect to policies focusing on the public research sector.

5.7 Extensions and Robustness Checks

In this section, we summarize the robustness of our estimation results and policy analysis to a number of variations on our sample and model. We provide in Appendix G the implied parameter estimates, the match between the model and the data moments, the social planner’s allocation, and the allocations that result from the different policy tools.

³¹We also look at policies in terms of the optimal number of labs U . We find that the social planner would decrease the number of labs, from a baseline value of $U = 0.49$ to an optimal value of $U = 0.20$, yielding welfare gains of 0.23%. The intuition is that because of the high fixed cost the planner will find it optimal to consolidate labs, even at the expense of moving up the variable cost curve.

³²The literature models the distinction between private and public research in terms of creative control (Lacetera, 2009; Howitt, 2000; Aghion and Tirole, 1997). The public sector, due to its non-profit nature, can essentially commit to leave scientists to pursue their own interests. Consequently if scientists value creative control they will have to be paid a wage premium in order to give it up. This feature has received empirical support in work by Stern (2004), who studies the job market for recent PhDs in biology. By using information on multiple job offers, he shows that wages are substantially lower in jobs that promise scientists a greater degree of freedom and disclosure of research.

Heterogeneous industries and firm types We re-estimate and evaluate the policy implications of extensions of the model featuring (i) industry heterogeneity and (ii) persistent firm types. To address concerns about industry heterogeneity, we split the economy into two groups of industries that are differentiated by their parameters governing basic research costs and benefits. To address concerns about firm heterogeneity, we introduce a firm type, that is fully persistent, and that determines a firm's basic research cost structure. The implications for policy, reported in Tables 6 and 9 of Appendix G.1 and G.2, are robust across these extensions.

Research types and institutions We also re-estimate two versions of the model without the distinction between research efforts: (i) we eliminate private basic research but still maintain public basic research; (ii) we eliminate basic research altogether, both private and public, and obtain a model with a single type of research. The estimates and policy implications are reported in Tables 12 and 15 of Appendix G.3. It turns out that the existence of investment in basic research by the private and public sectors is important for our policy conclusions. Without these distinctions in research investment, and their associated spillovers, optimal policy would actually discourage research relative to the baseline economy.

Limited diffusion of private basic research We then consider two factors related to public basic research. First, research produced in the public sector is a public good, while private research may be kept secret by firms. Therefore, we consider a version of the model in which private basic research still benefits the firm but makes only a limited contribution to future economic growth. That is, firms keep a fraction of their research findings secret, and, hence, future innovators cannot build on that knowledge. The policy implications are depicted in Figure 2 of Appendix G.4.

Spillovers from applied research and CRRA parameter As a final set of robustness checks, we first consider whether applied research is necessarily targeted to specific product lines/industries. We extend the cross-industry spillover to applied research as well and provide a quantification of the model fit as a function of different applied step sizes in Figure 3 of Appendix G.5. Second, since the identification of the Constant Relative Risk Aversion (CRRA) parameter is indirect, we also perform in Table 19 of Appendix G.6, a robustness check in which we fix the parameter to a standard value of the literature and re-estimate the model.

6 Conclusion

In this paper, we distinguished between basic and applied research investment and identified the spillovers associated with each. We built a new micro-founded model of endogenous growth with two types of research investment (basic and applied) and two types of entities (private firms and public research labs). Our quantitative analysis highlighted the importance of these distinctions. First,

in the competitive equilibrium, there is an overinvestment in applied research and underinvestment in basic research. As a result, imposing a uniform research subsidy does not generate the expected welfare improvement due to inefficient cross-subsidization of applied research. A key message of our paper is, therefore, that such policies can accentuate the dynamic misallocation in the economy. Second, there are important complementarities between public and private research investments. When public research labs produce more basic knowledge, private firms build on this and produce more impactful applied innovations.

Overall, our analysis relative to the uniform research subsidy resembles a reform of the French R&D policy. In 2008, the French government introduced a 30% tax credit for all of firms' R&D related expenditures. While the new system represented a significant subsidy to private R&D, with an annual budget of five billion Euros, its effectiveness in boosting innovation has been widely criticized (Larousserie, 2015).

Our findings can account for the limited impact of such policies and to the need for targeting basic research more directly. Our paper took a first step in trying to quantify the inefficiencies regarding different types of research and innovation efforts. There are still important open questions awaiting further study. Amongst them, our analysis highlighted the need for a better understanding of the public research sector, the design of the patent system, and the incorporation of firm heterogeneity and non-R&D firms. We hope that further structural work will be undertaken to enhance our understanding of the aforementioned issues, which can then guide the relevant innovation policies.

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